Hypothesis testing

DANIELA HUPPENKOTHEN

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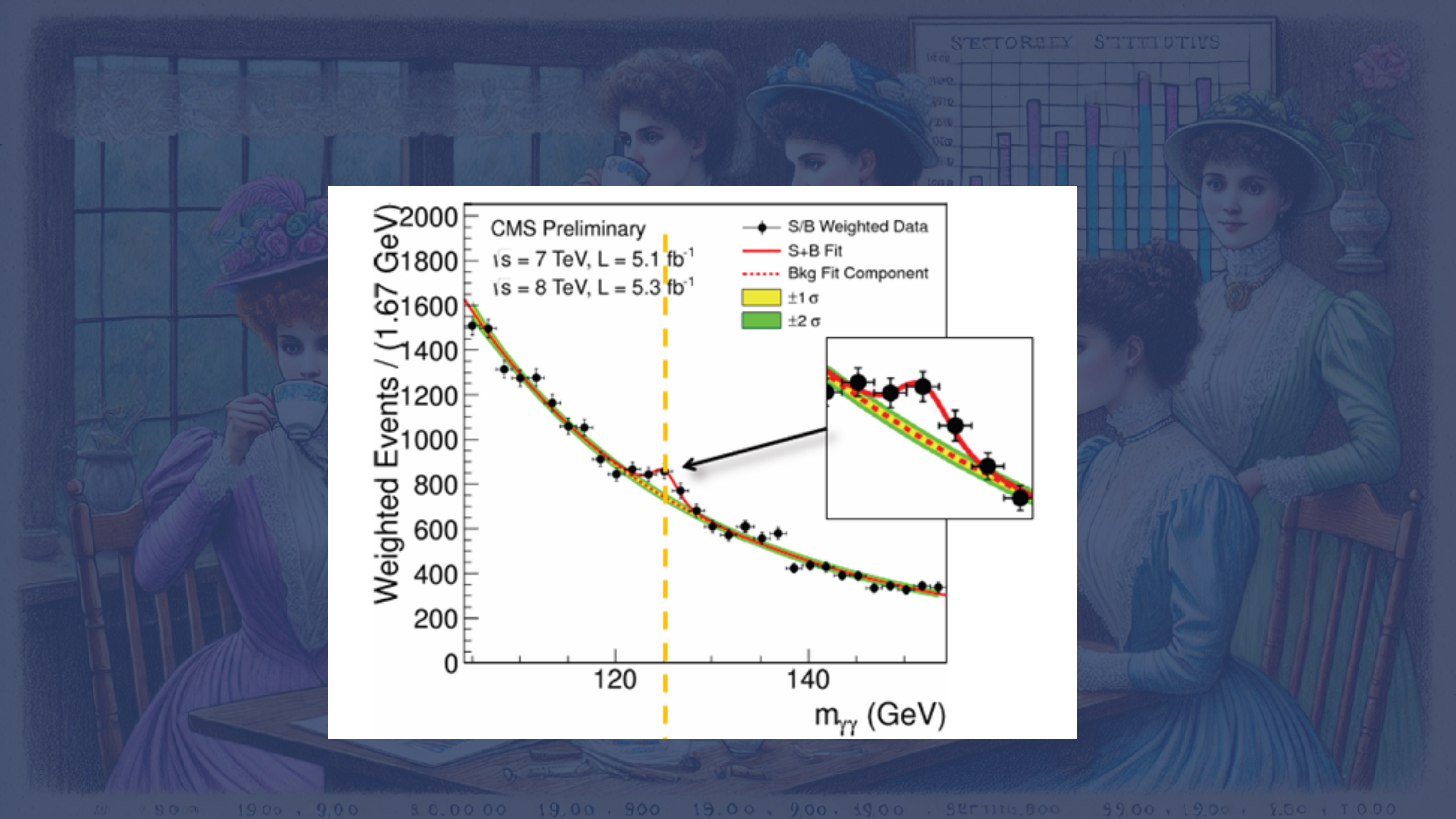
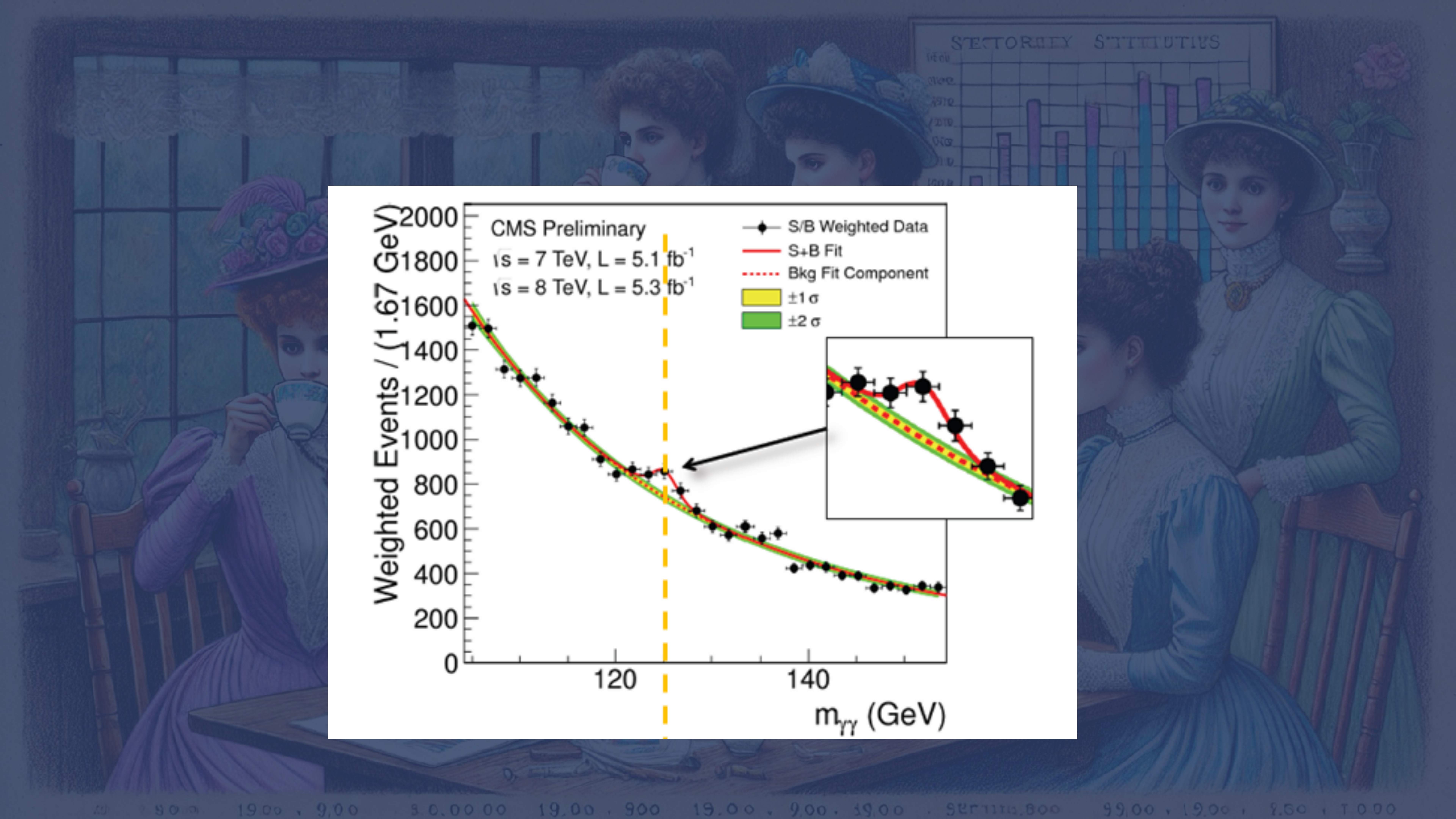

CMS Preliminary
$\sqrt{s}$ = 7 TeV, L = 5.1 fb^{-1}
$\sqrt{s}$ = 8 TeV, L = 5.3 fb^{-1}
S/B Weighted Data
S+B Fit
Bkg Fit Component
±1 σ
±2 σ
Weighted Events / (1.67 GeV)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
120
140
$m_{\gamma\gamma}$ (GeV)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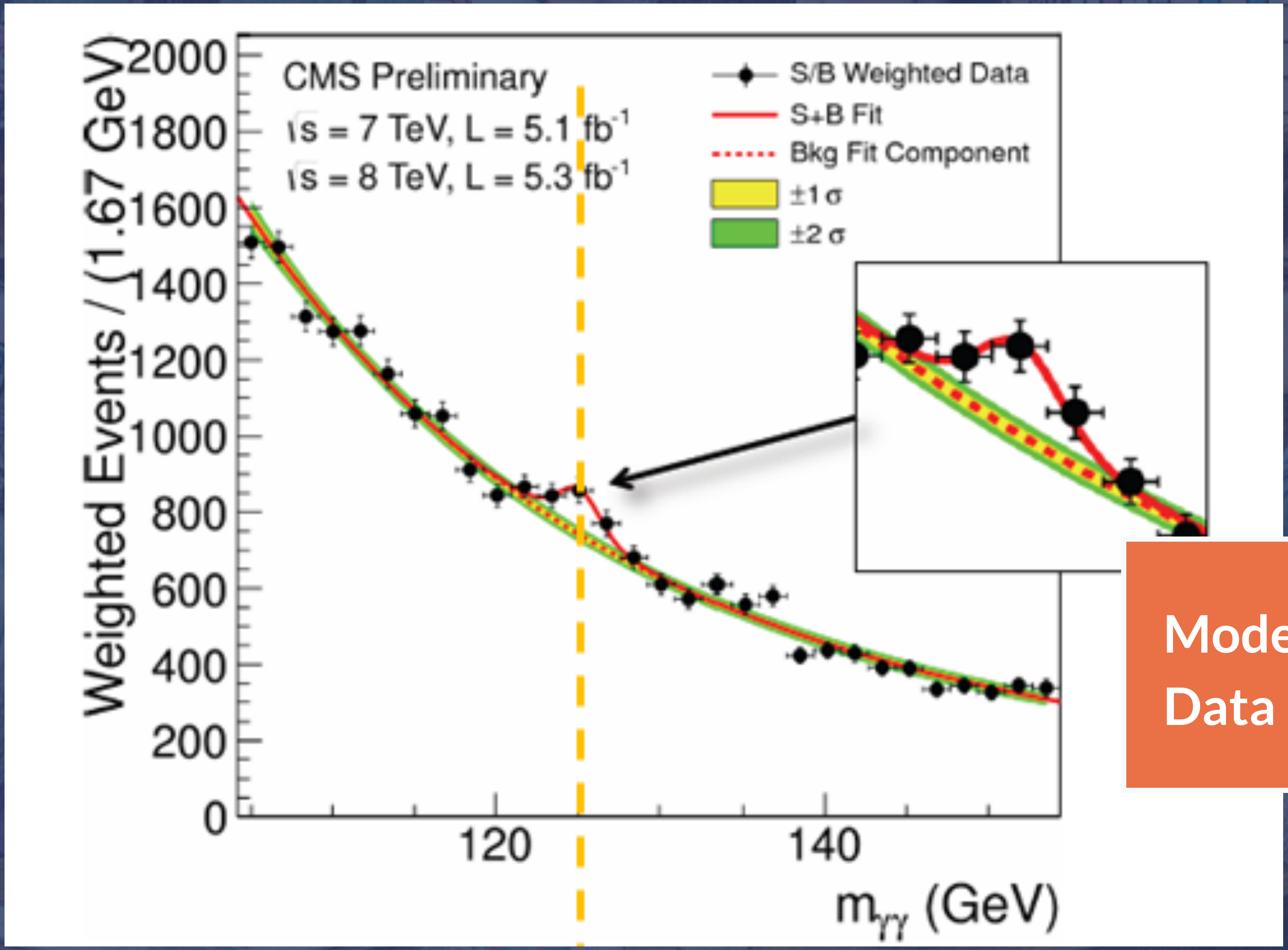
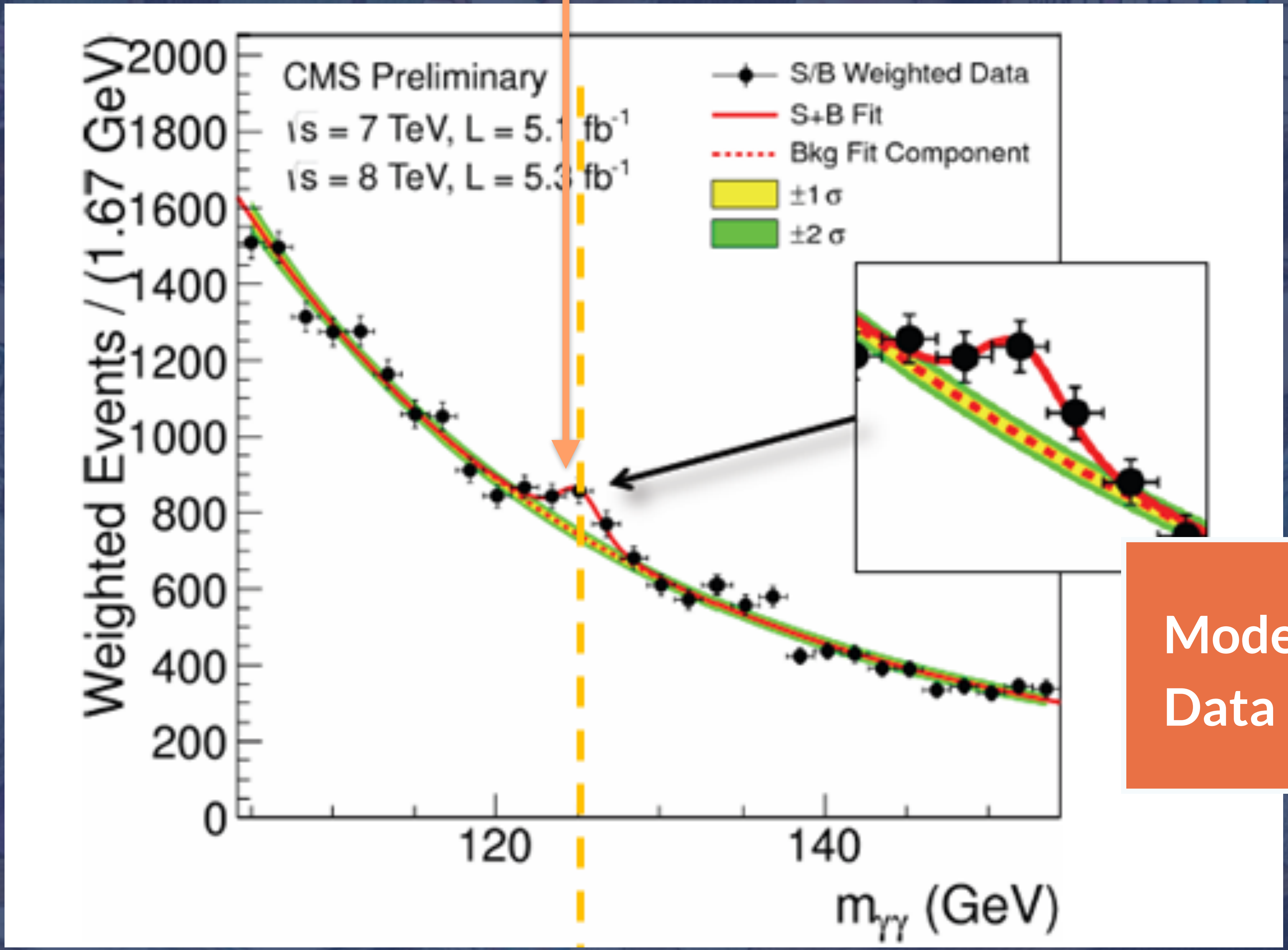


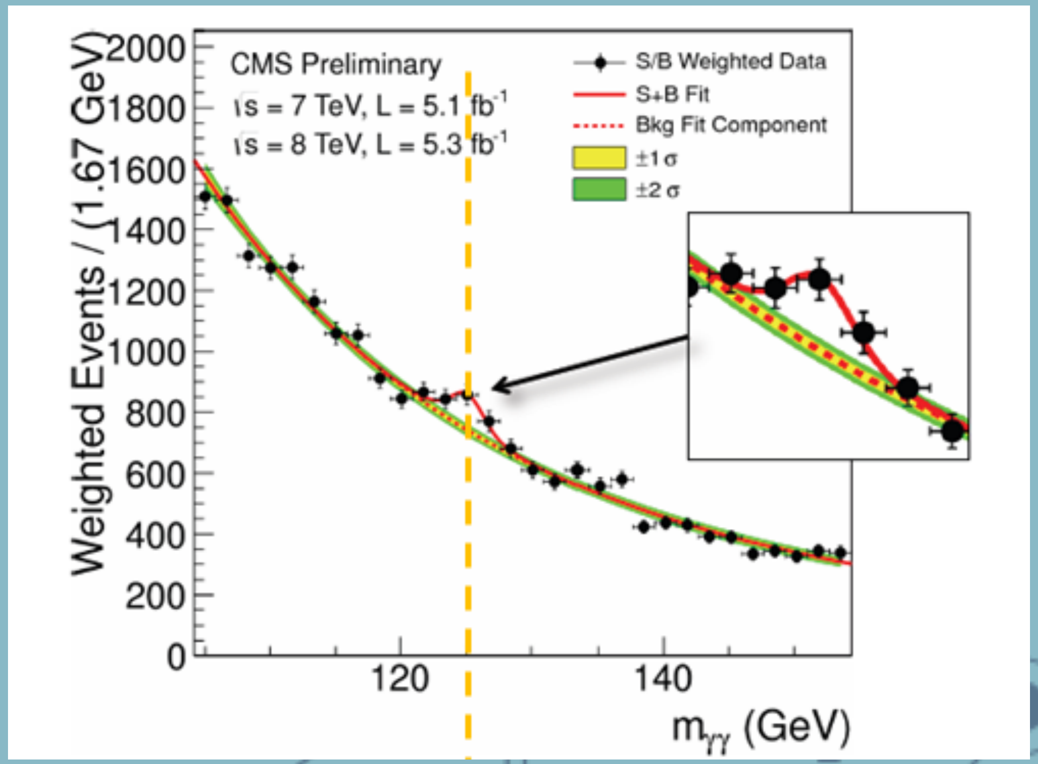
Model misspecification?
Data issues?

Is there a new particle in the CMS data?



Model misspecification?
Data issues?

Brainstorm: How can we figure out whether there's a particle in the data?



- What do we need to find out from the data to tell whether there's a new particle?
- What are the random variables? Are there quantities we could turn into random variables?
- Can you tie these to particular distributions or statistical concepts that might be helpful?

Hypothesis Testing

Exercise: Finding planets in light curves

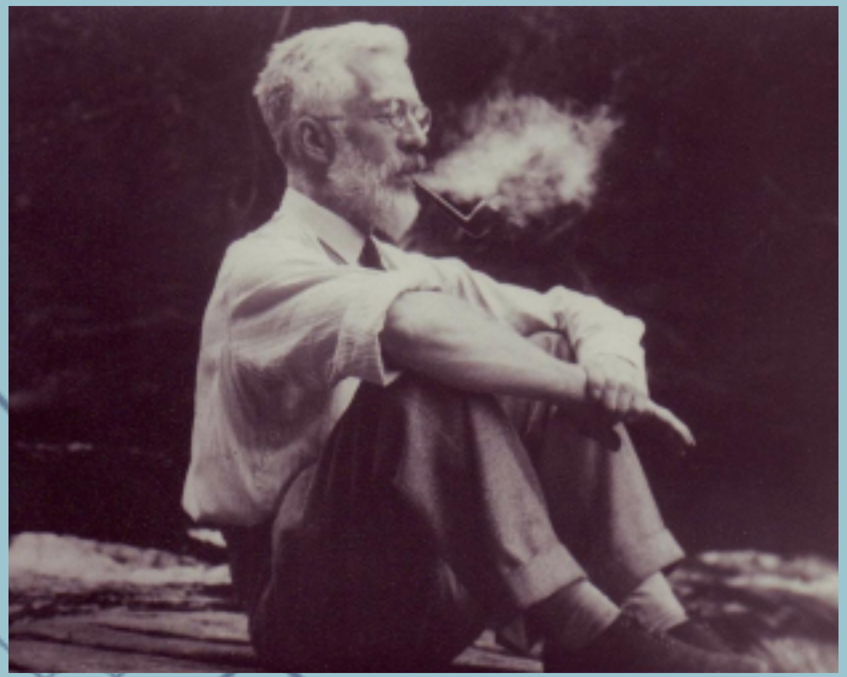
Hypothesis versus significance testing

Hypothesis versus significance testing

Hypothesis versus significance testing

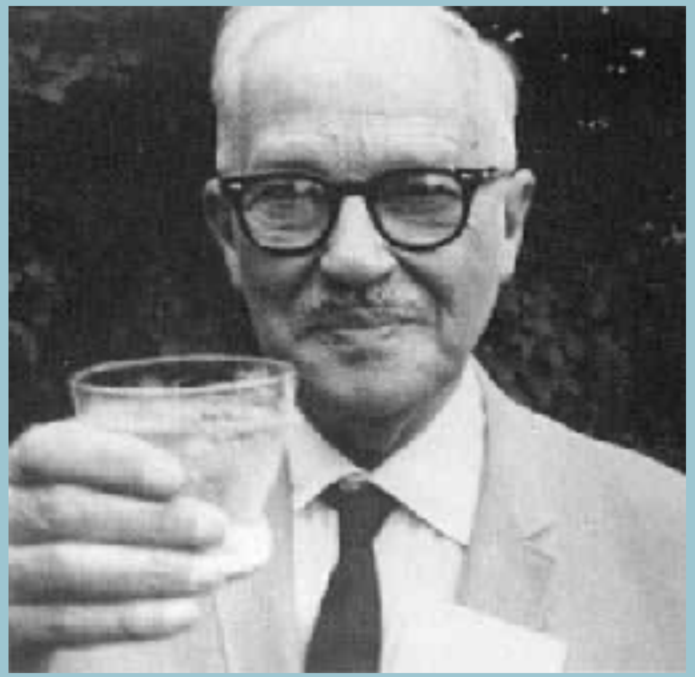
<u>Significance test (R.A. Fisher)</u>: **the p-value denotes the degree of significance, i.e. how extreme the observations are under the null hypothesis**

<u>Hypothesis testing (J. Neyman + E. Pearson)</u>: set up both null hypothesis and alternative hypothesis, and a decision rule: if $T > T_{\text{crit}}$, the result is "significant", choose T_{crit} such that you are wrong only an acceptably small number of times



R. A. Fisher

vs.



J. Neyman

Hypothesis versus significance testing

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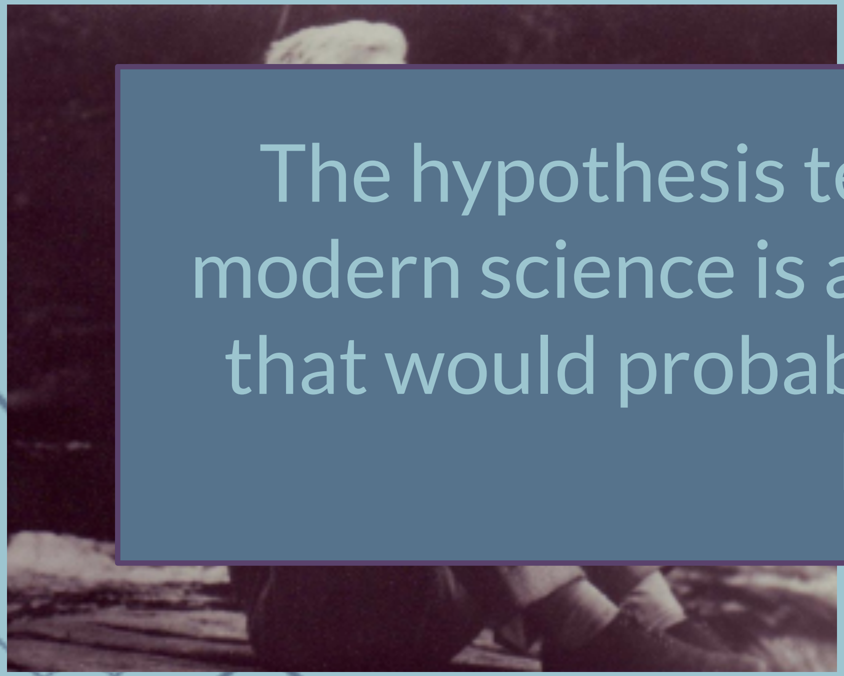
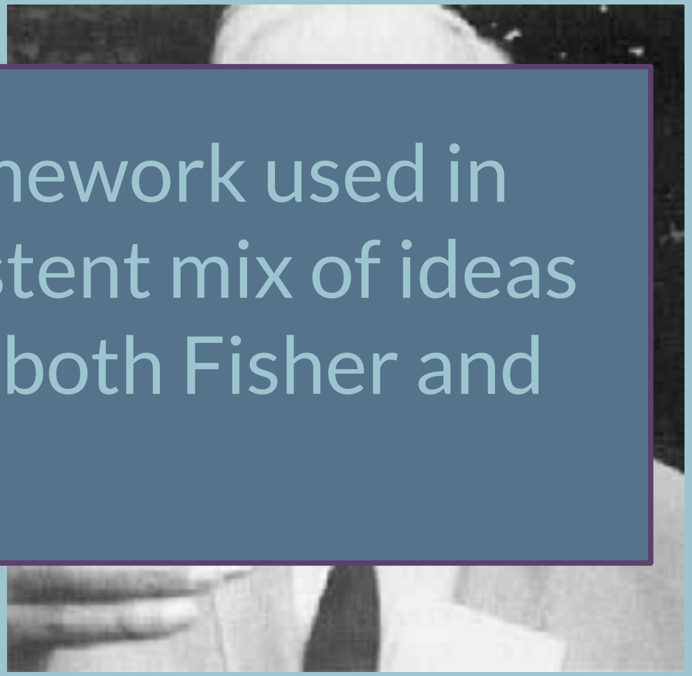




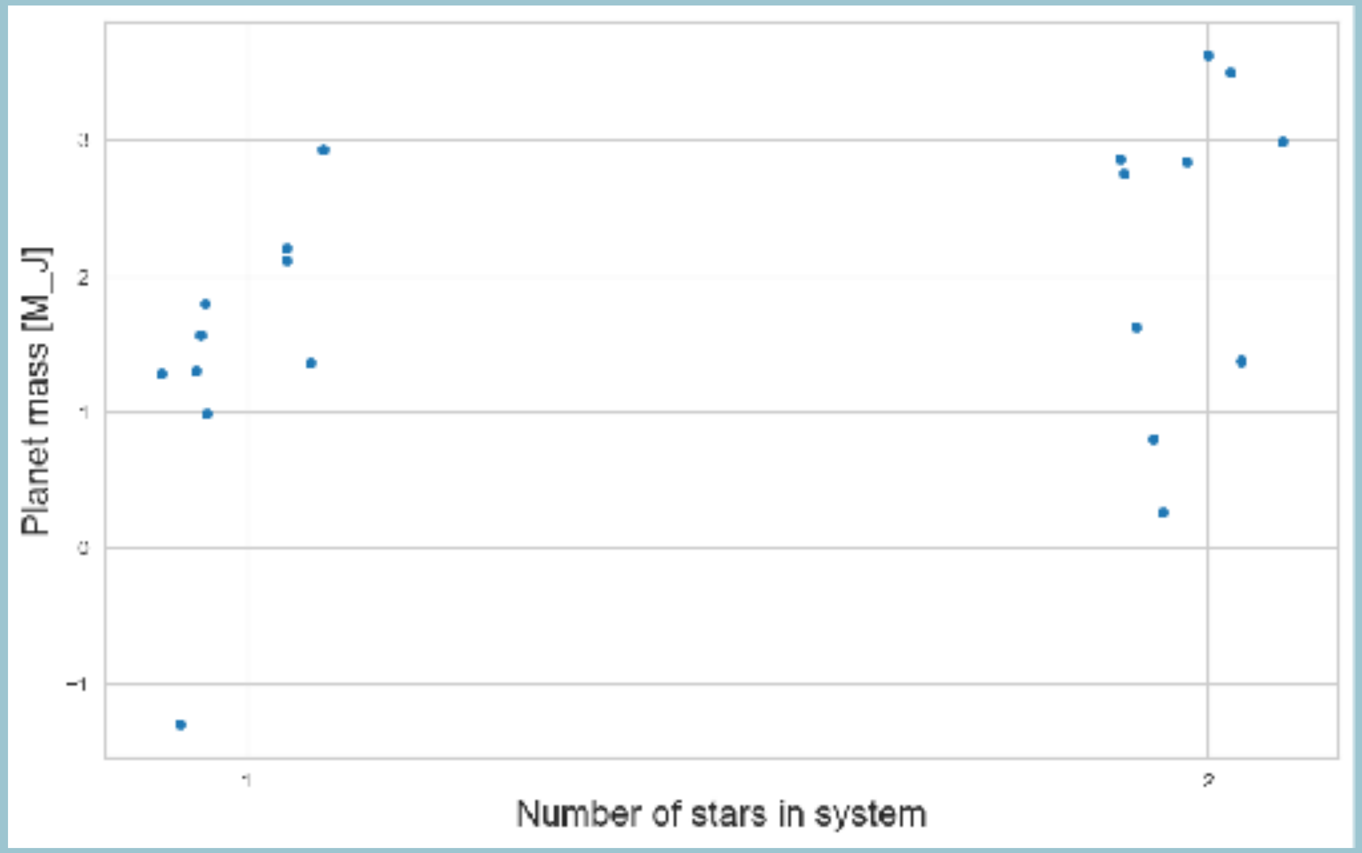
The hypothesis testing framework used in modern science is an inconsistent mix of ideas that would probably horrify both Fisher and Neyman

R. A. Fisher

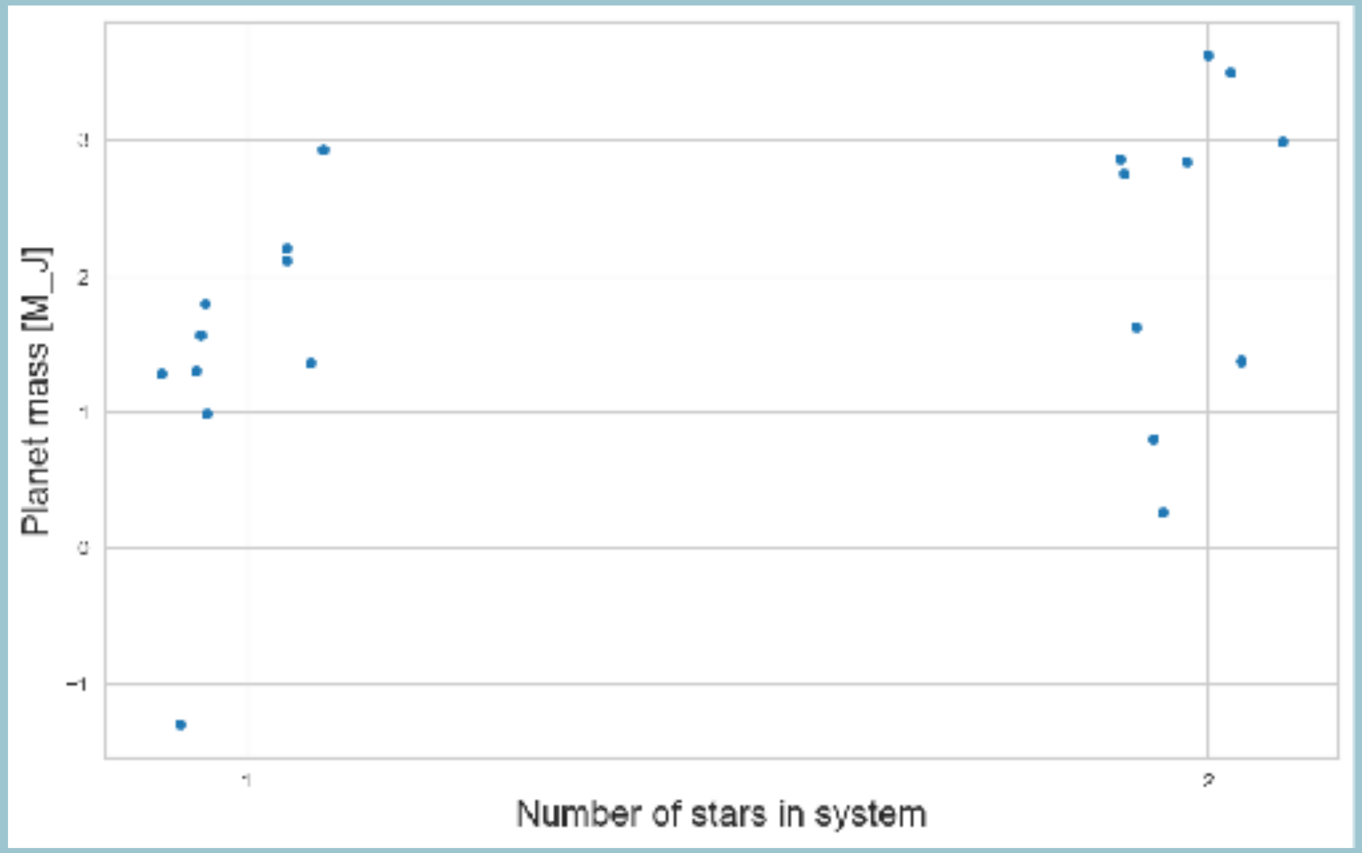
J. Neyman

A simpler problem: are planets around binary stars more massive than planets around single stars?

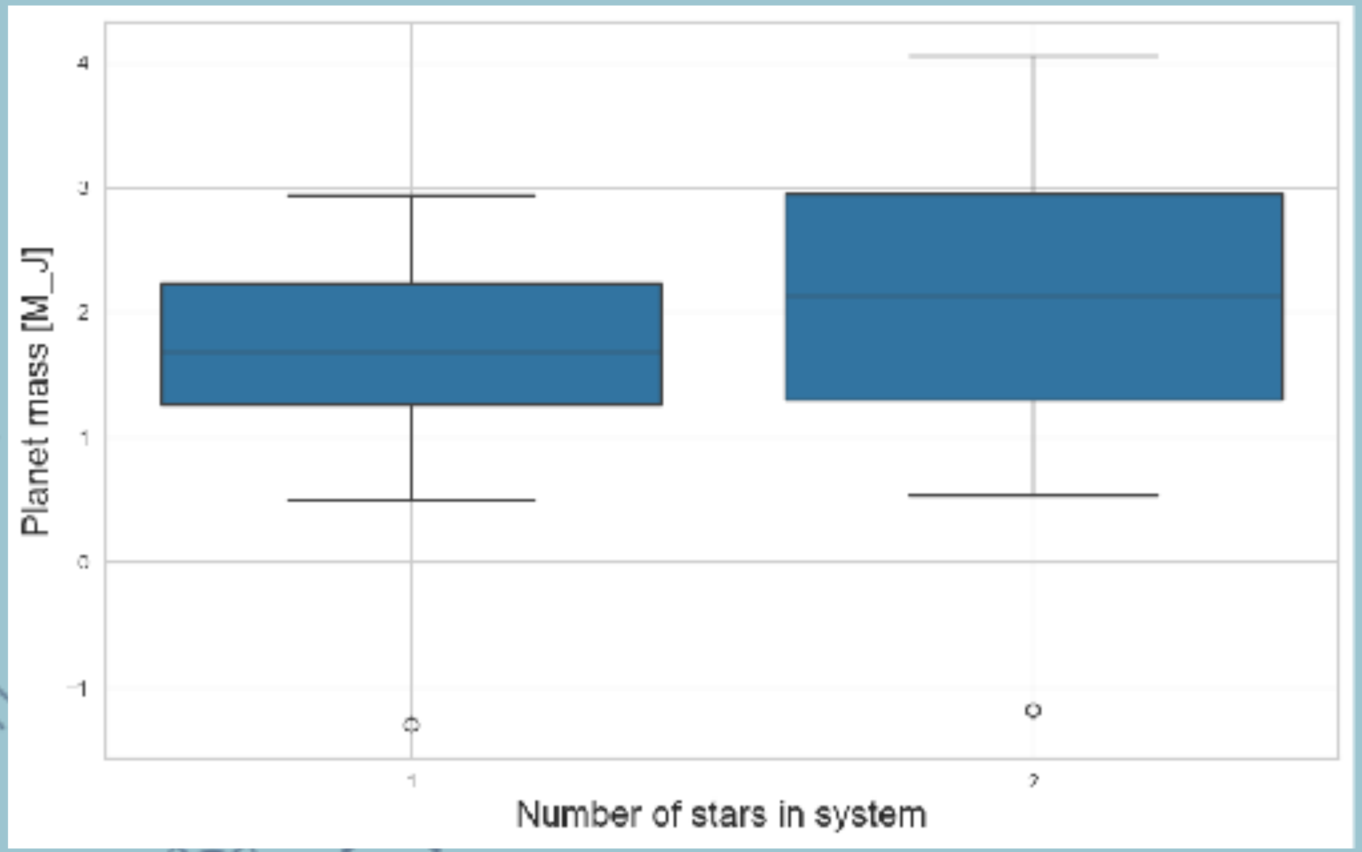
Are planets around binary stars more massive than planets around single stars?



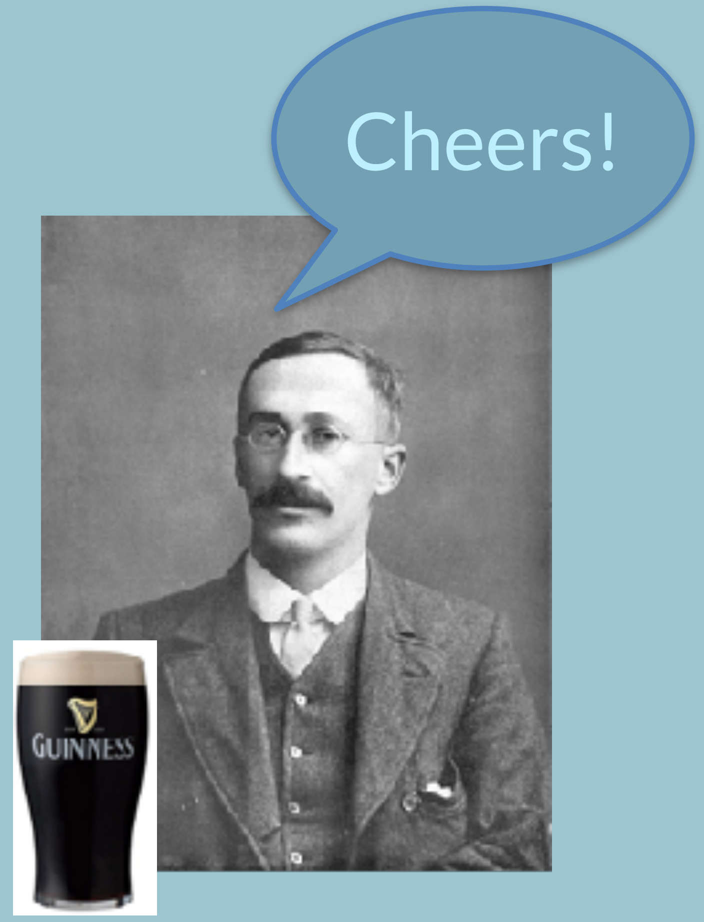
Are planets around binary stars more massive than planets around single stars?



Box plot:

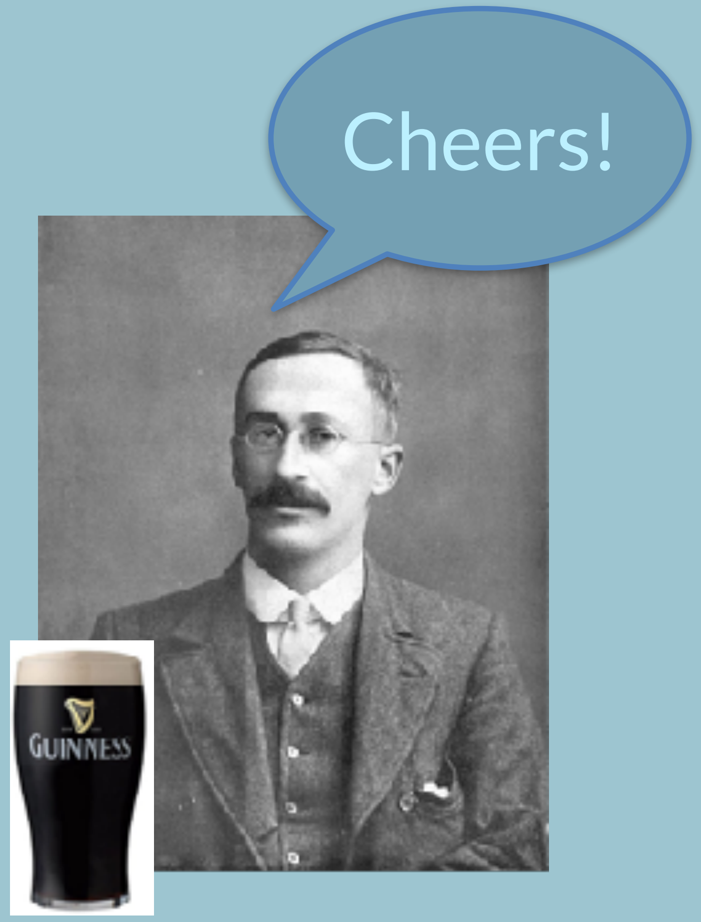


Now what?



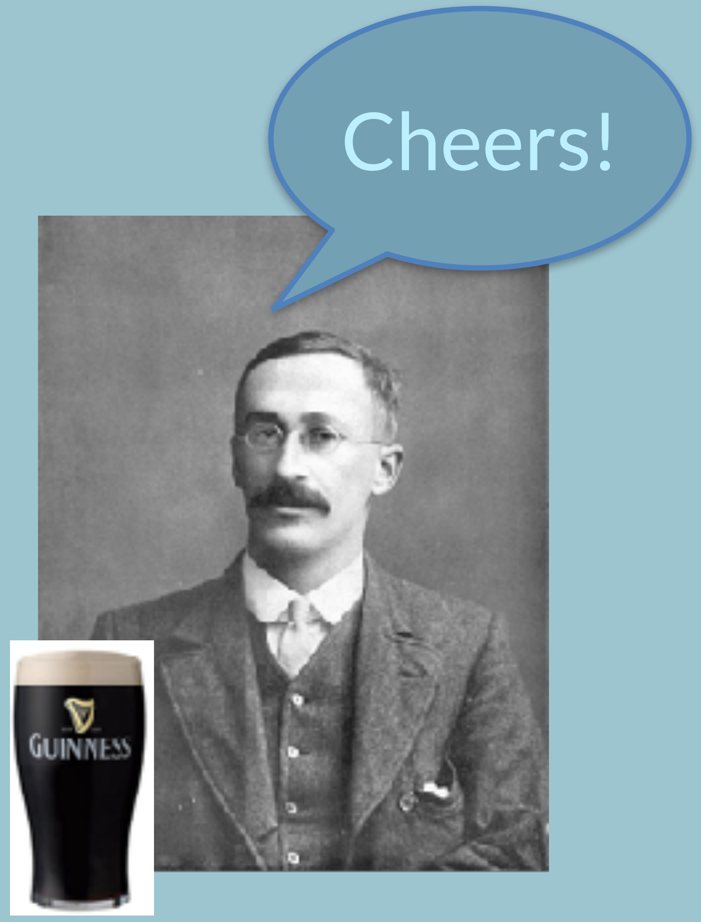
Now what?

- Null hypothesis:



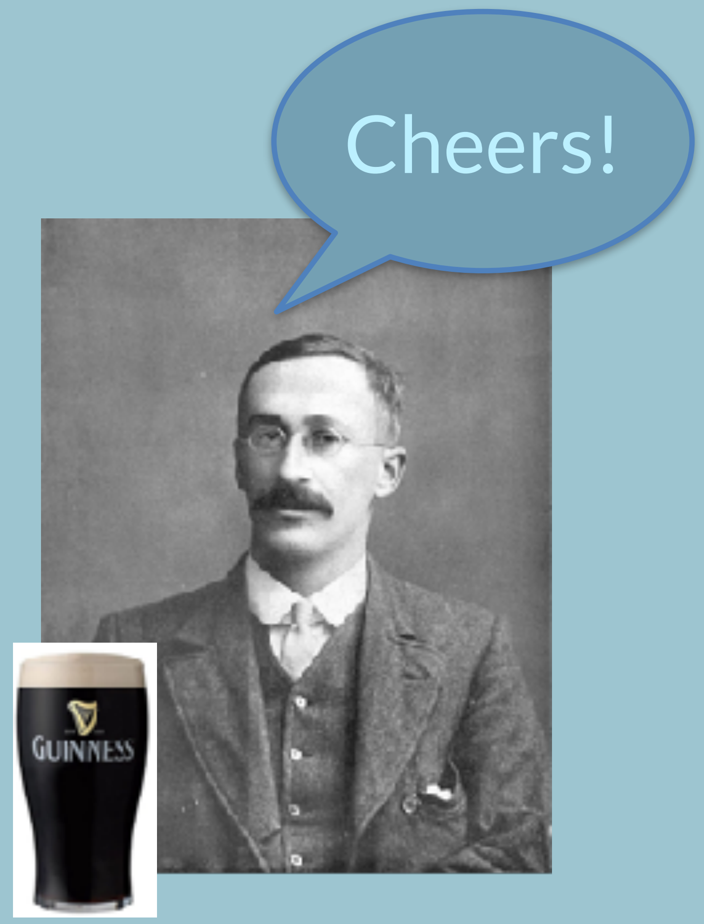
Now what?

- Null hypothesis:
- Alternative hypothesis:



Now what?

- Null hypothesis:
- Alternative hypothesis:
- Test statistic:



Now what?

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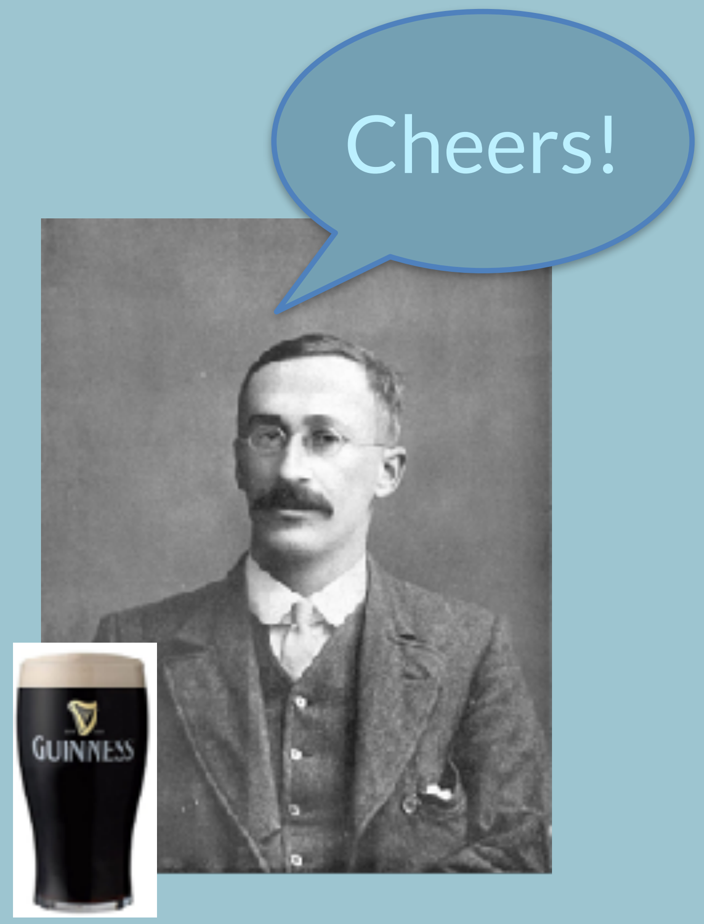
… and now we need a distribution!



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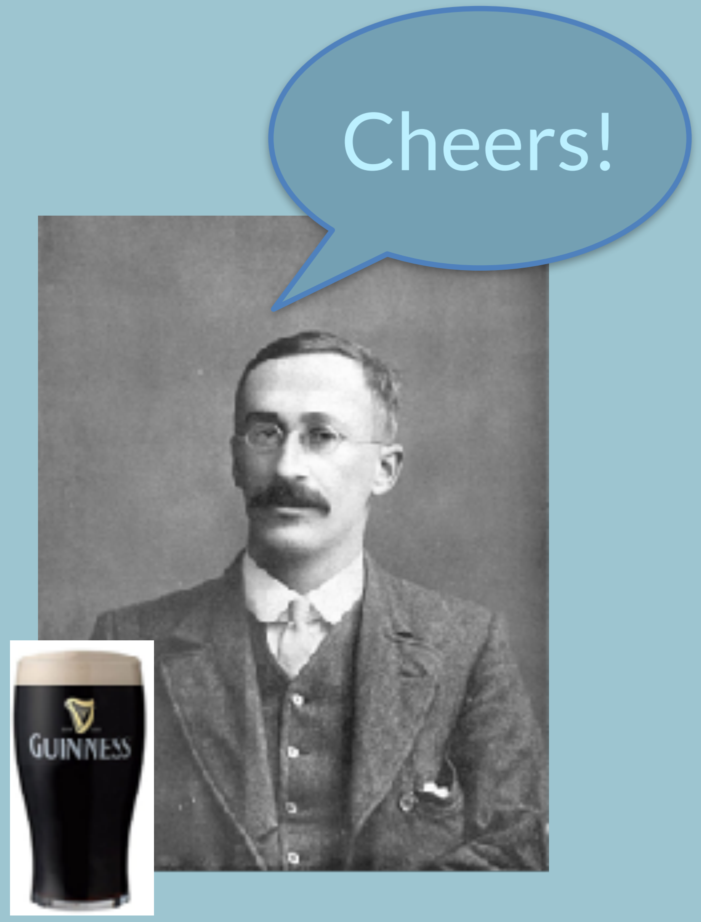

Now what?

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- Alternative hypothesis:
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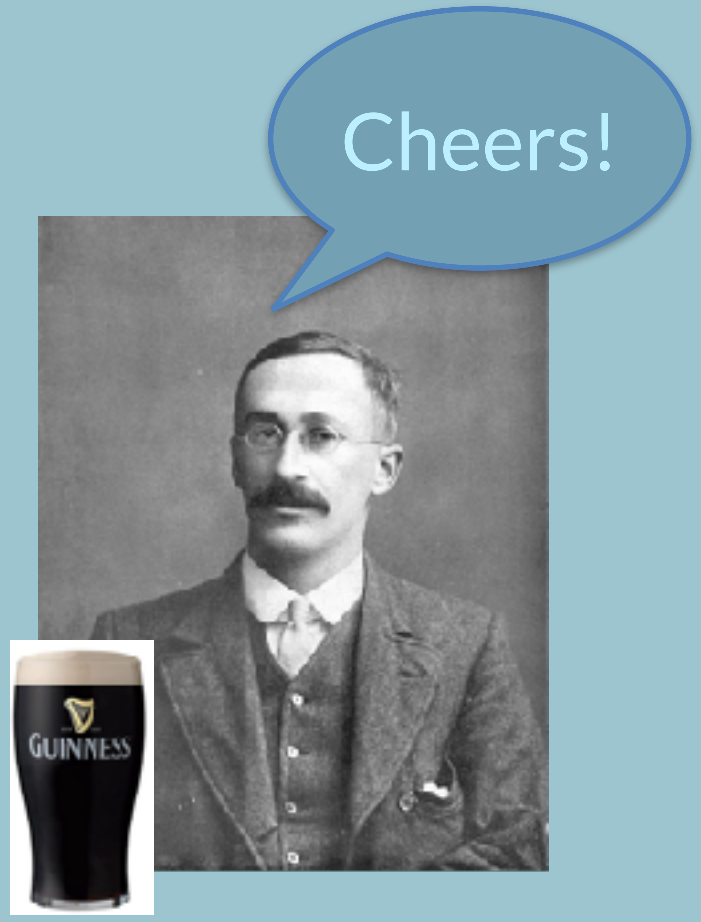
… and now we need a distribution!

- **Option 1**: simulate lots of time from the null hypothesis as we’ve done in the exercise, build distribution empirically



Now what?

- Null hypothesis:
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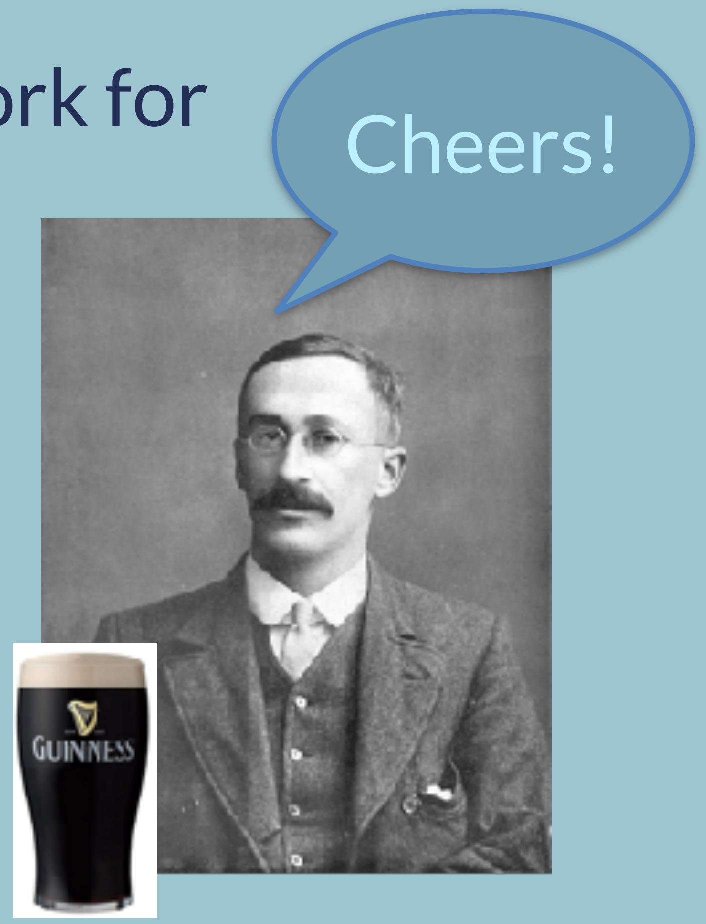
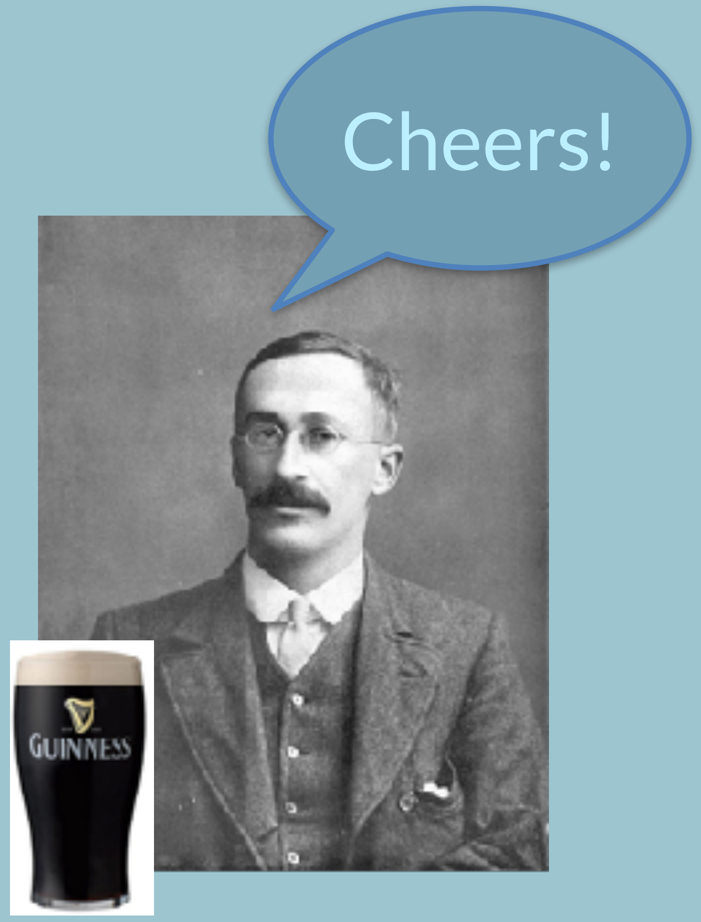
… and now we need a distribution!

- **Option 1**: simulate lots of time from the null hypothesis as we've done in the exercise, build distribution empirically
- **Option 2**: see if mathematicians have done the work for us

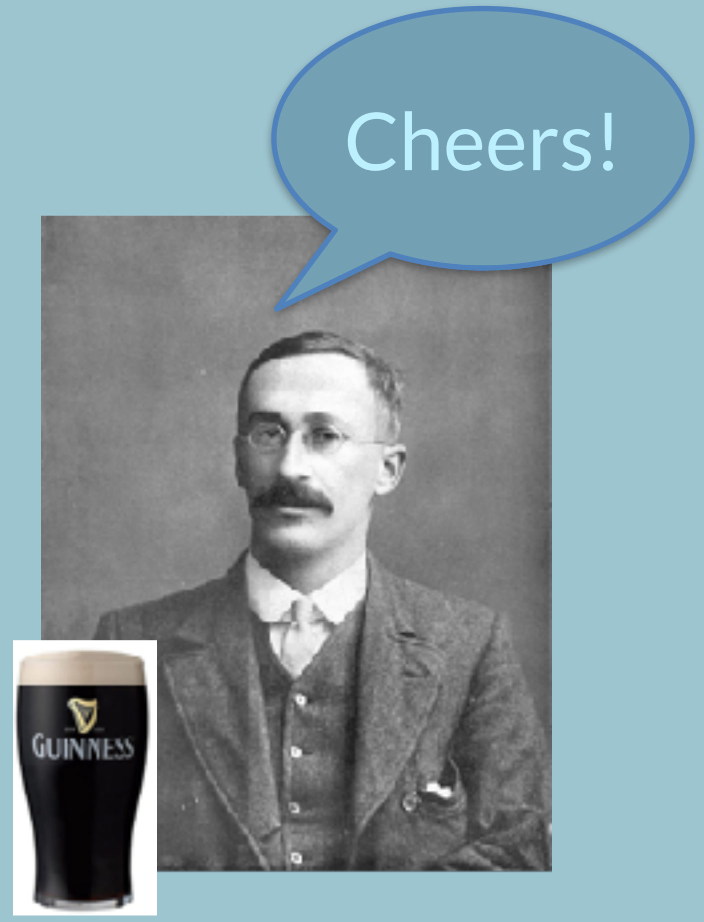


t-test assumptions:



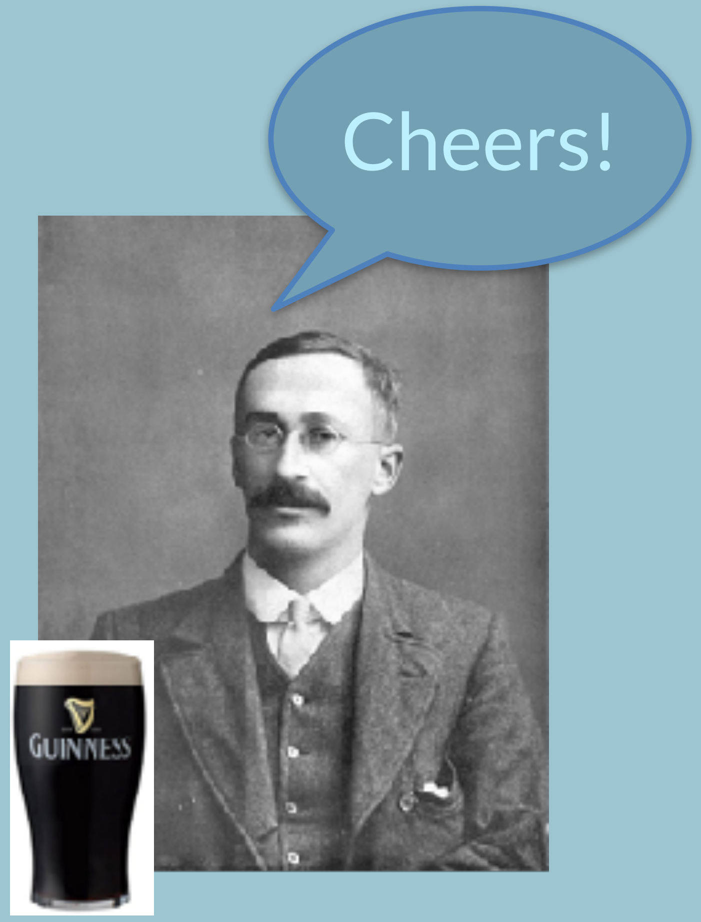
t-test assumptions:

- Data is a continuous variable



t-test assumptions:

- Data is a continuous variable
- Observations are independent, random samples from the population, without significant outliers



t-test assumptions:

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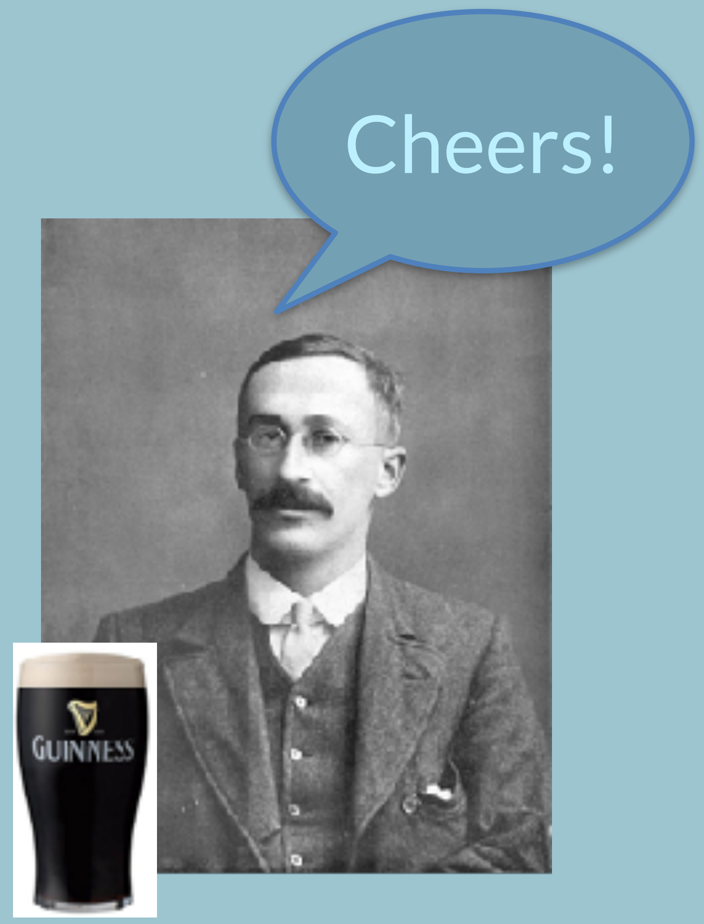
- Data is approximately normally distributed



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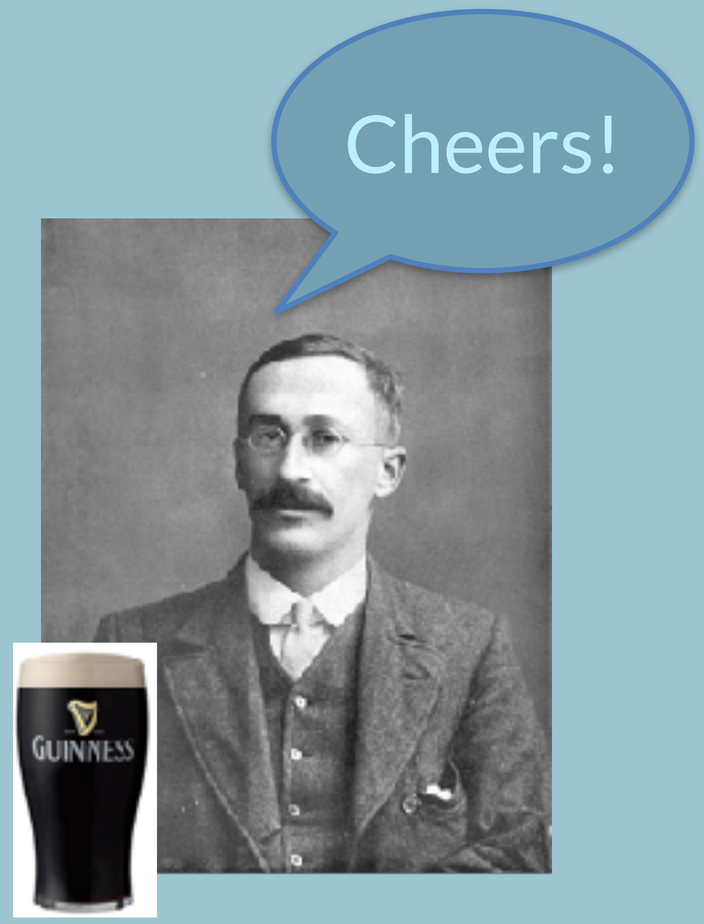
- Variances of the groups are equal



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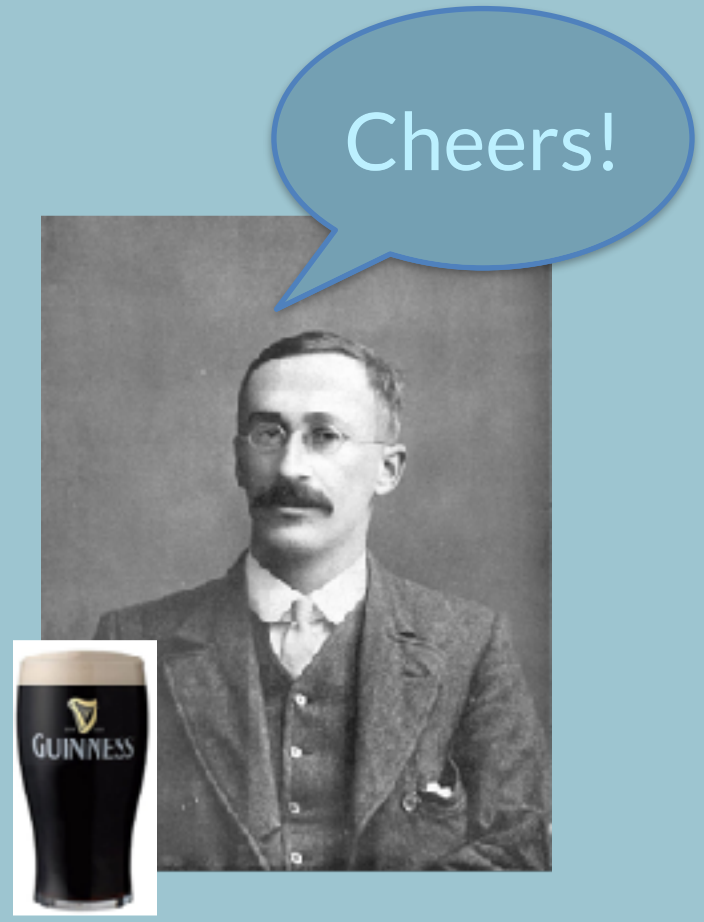
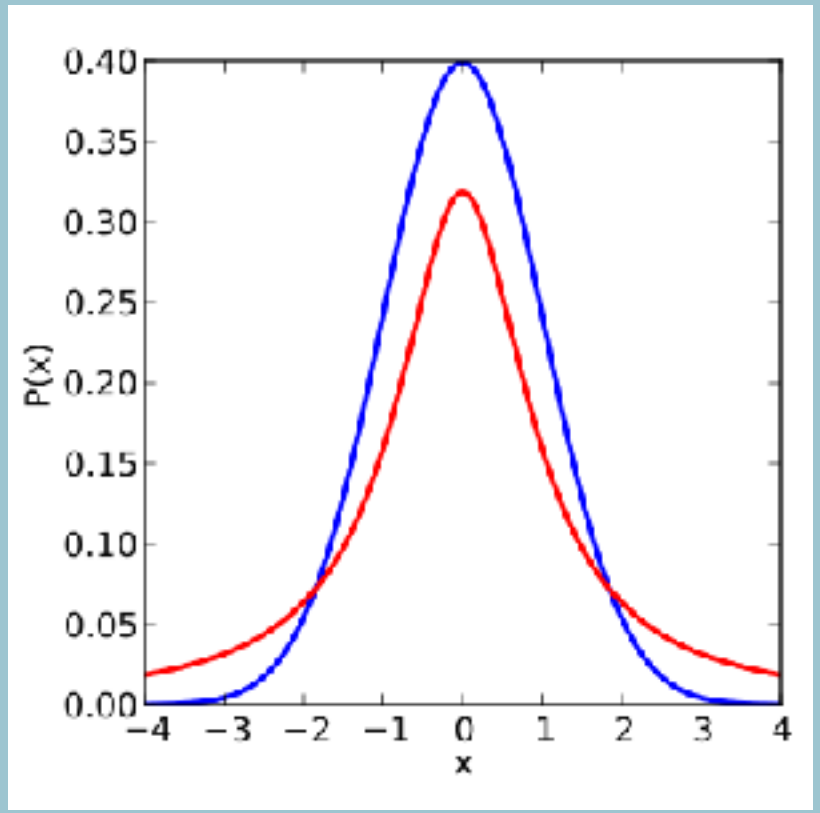
Is this true in our case?

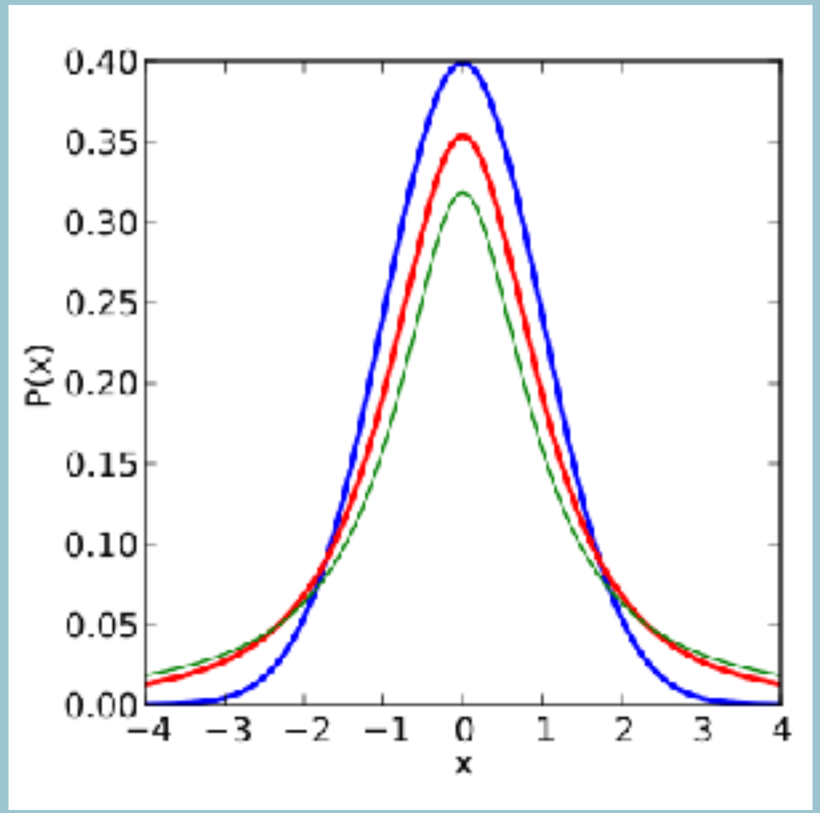


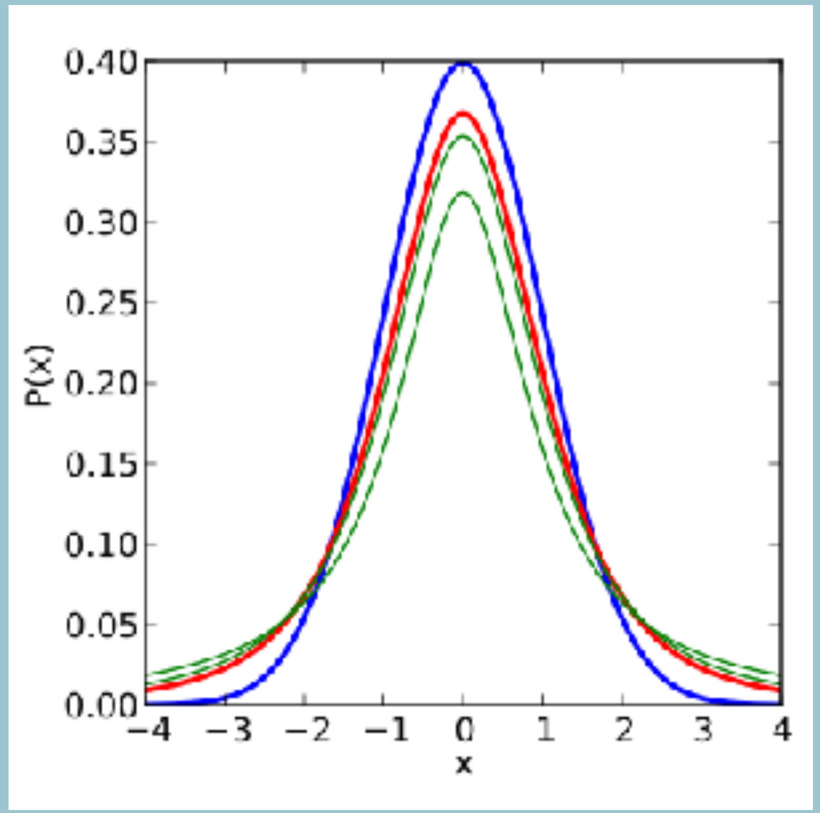
Student-t distribution

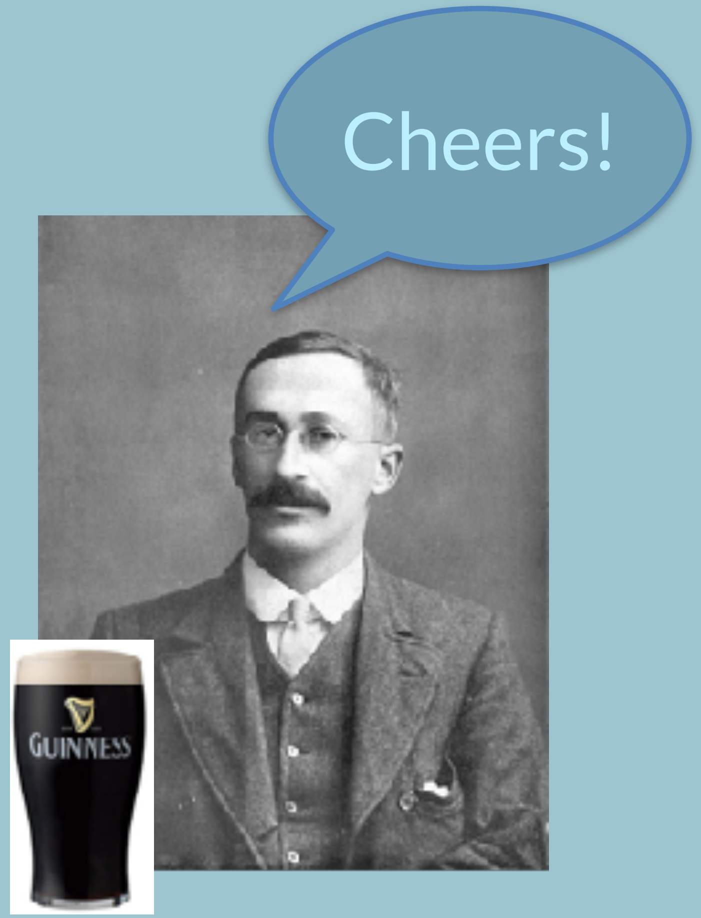
Red: student t

Blue: Gaussian









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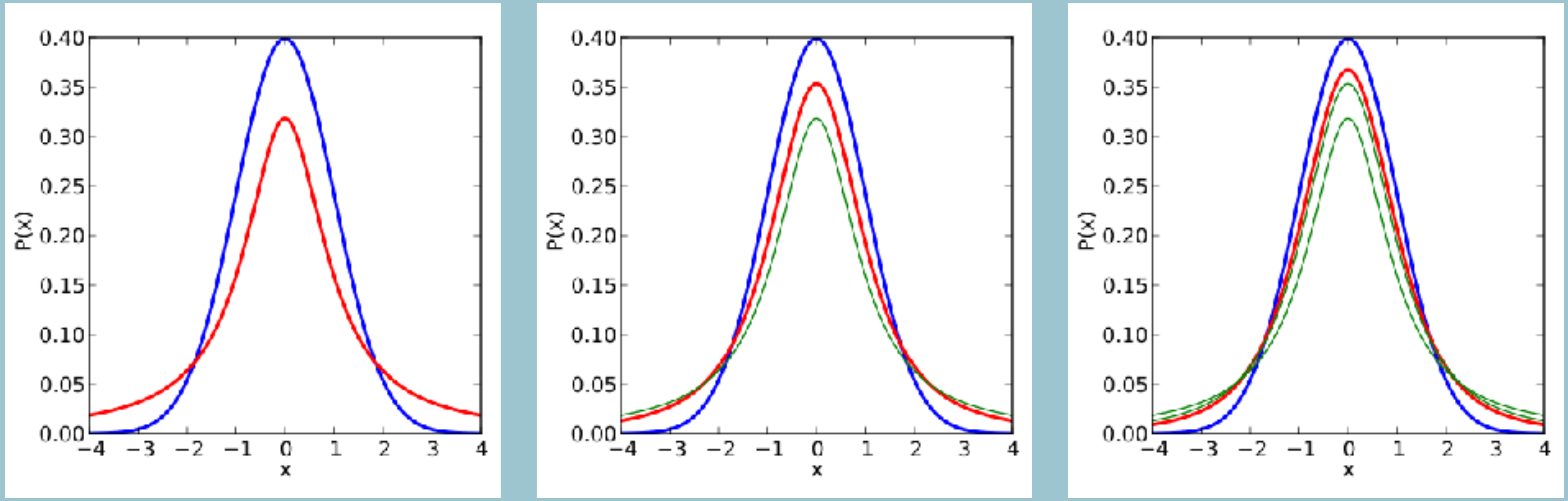


Under the null hypothesis (equal sample means), the test statistic follows a student-t distribution

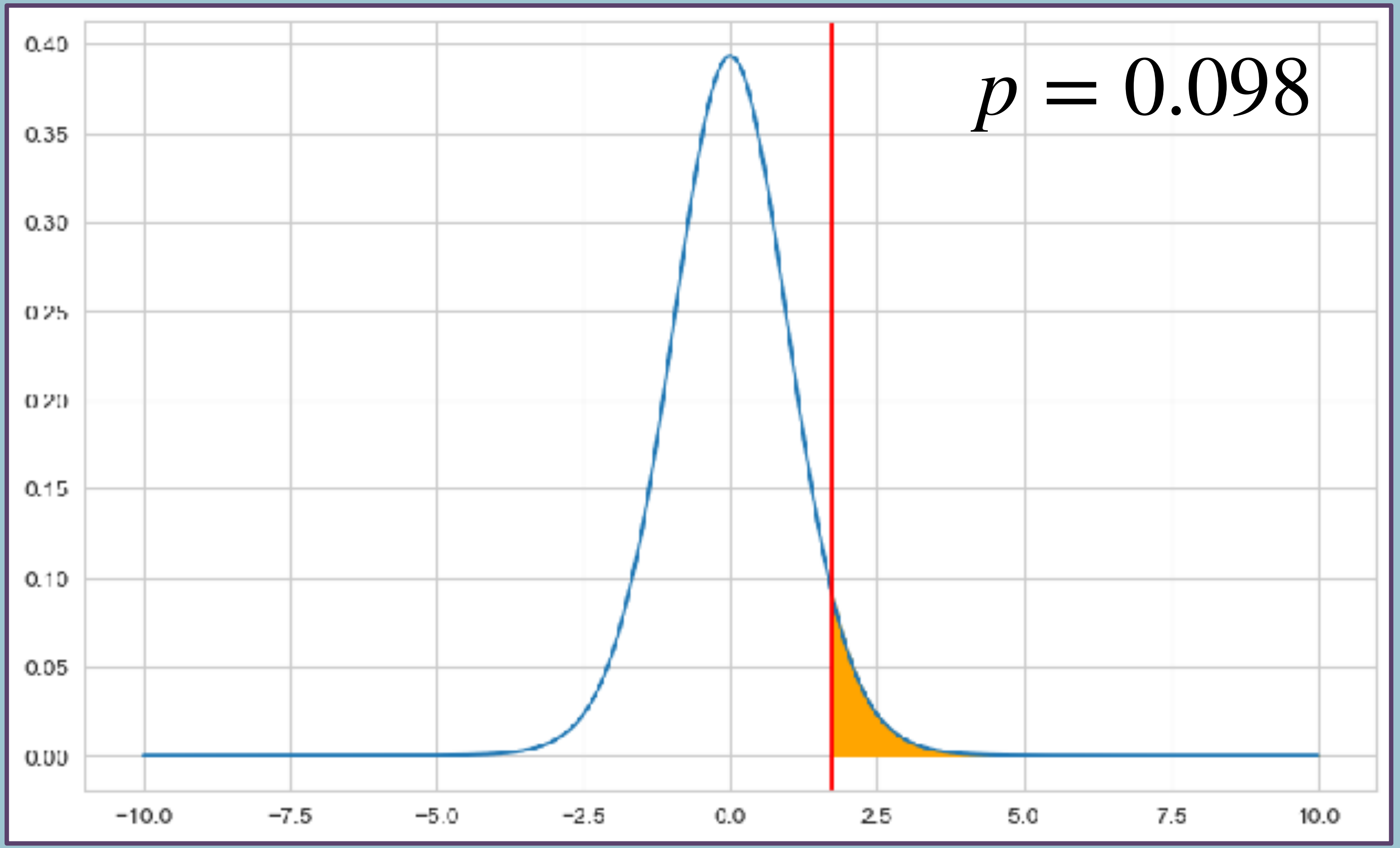
Cheers!

p-values

p-values

Under the null hypothesis, how probable is it to obtain a value of my test statistics at least as extreme or more extreme than the one I've observed?

p-values



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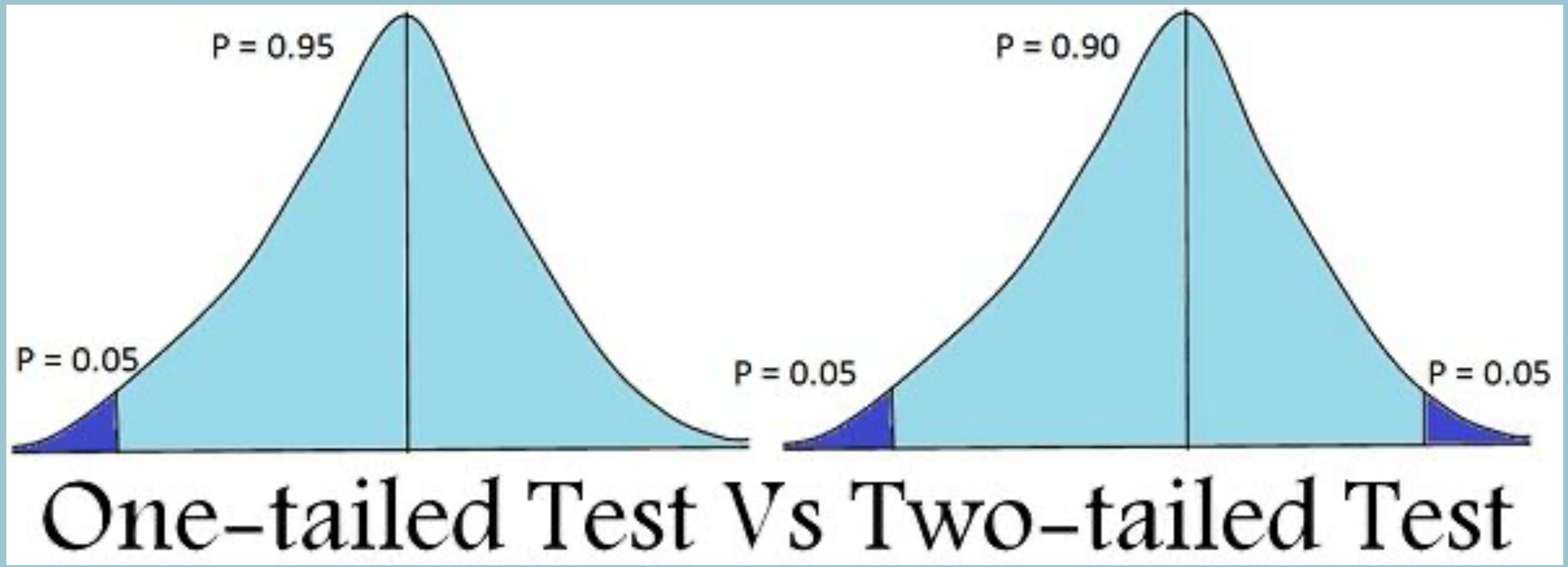
One-tailed versus two-tailed tests

https://rpsychologist.com/d3/nhst/



One-tailed versus two-tailed tests

- Decide on acceptable false positives versus false negatives

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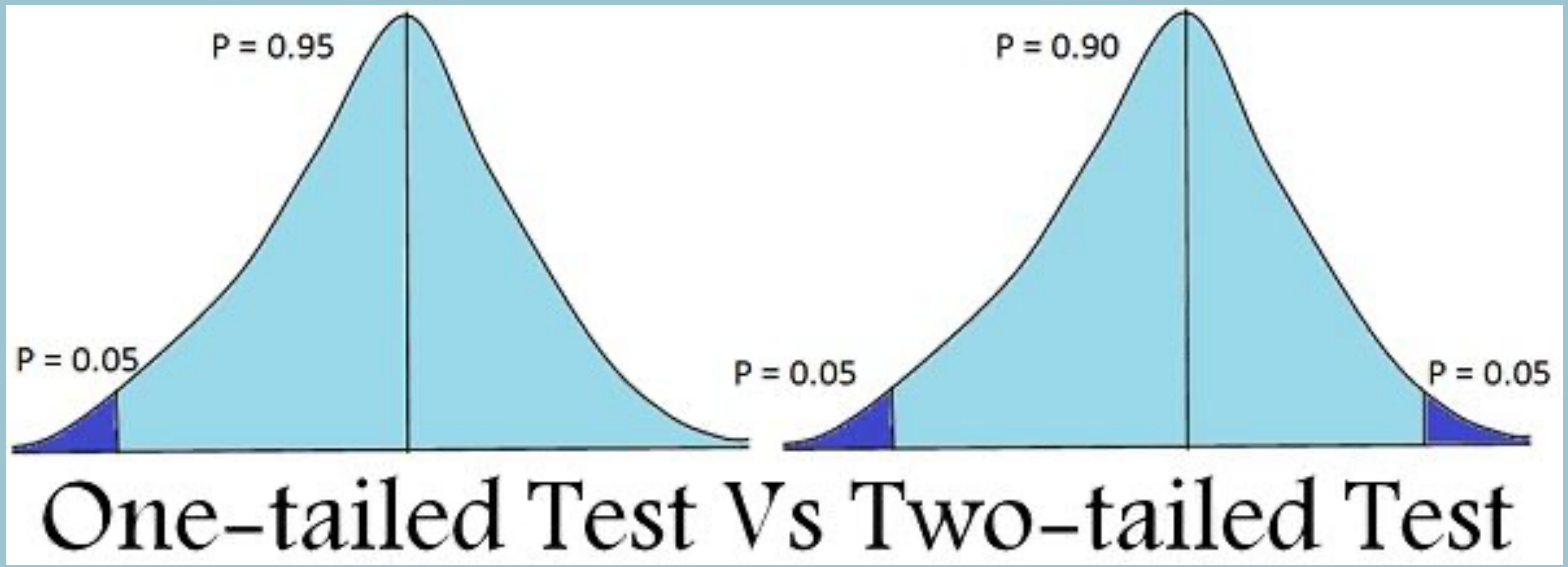

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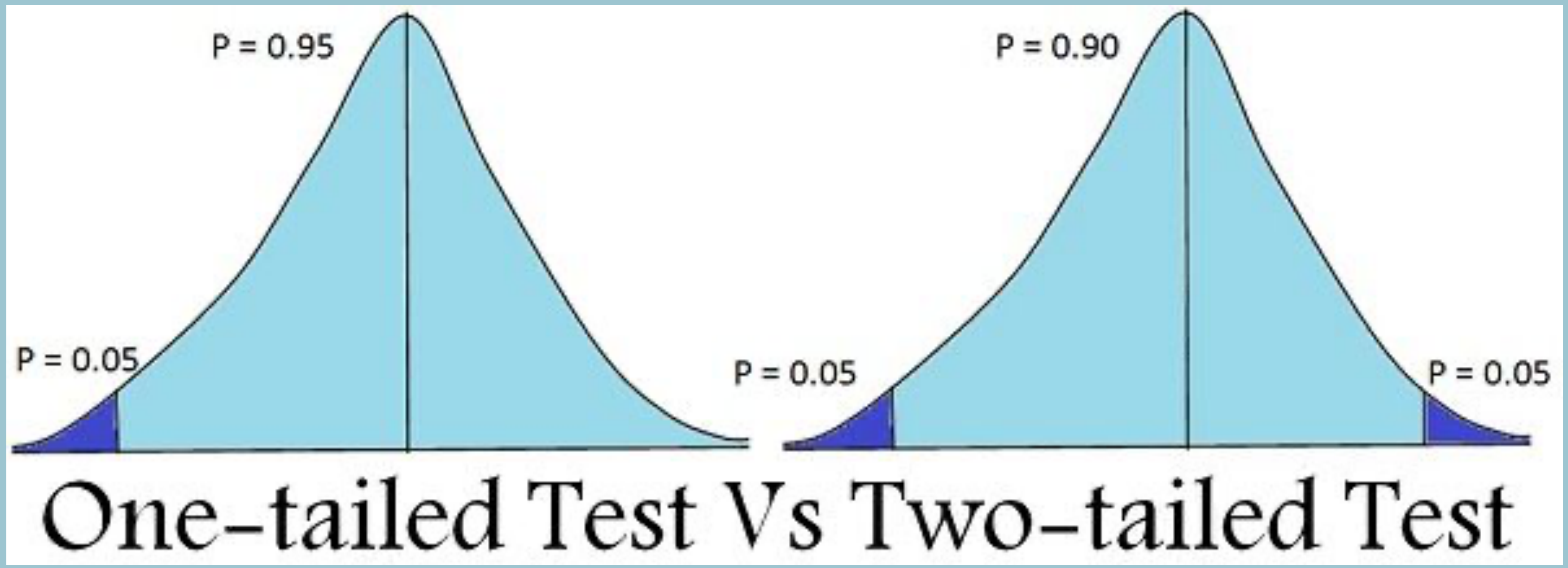

One-tailed versus two-tailed tests

Do we need a one-tailed or two-tailed test here?

- Decide on acceptable false positives versus false negatives

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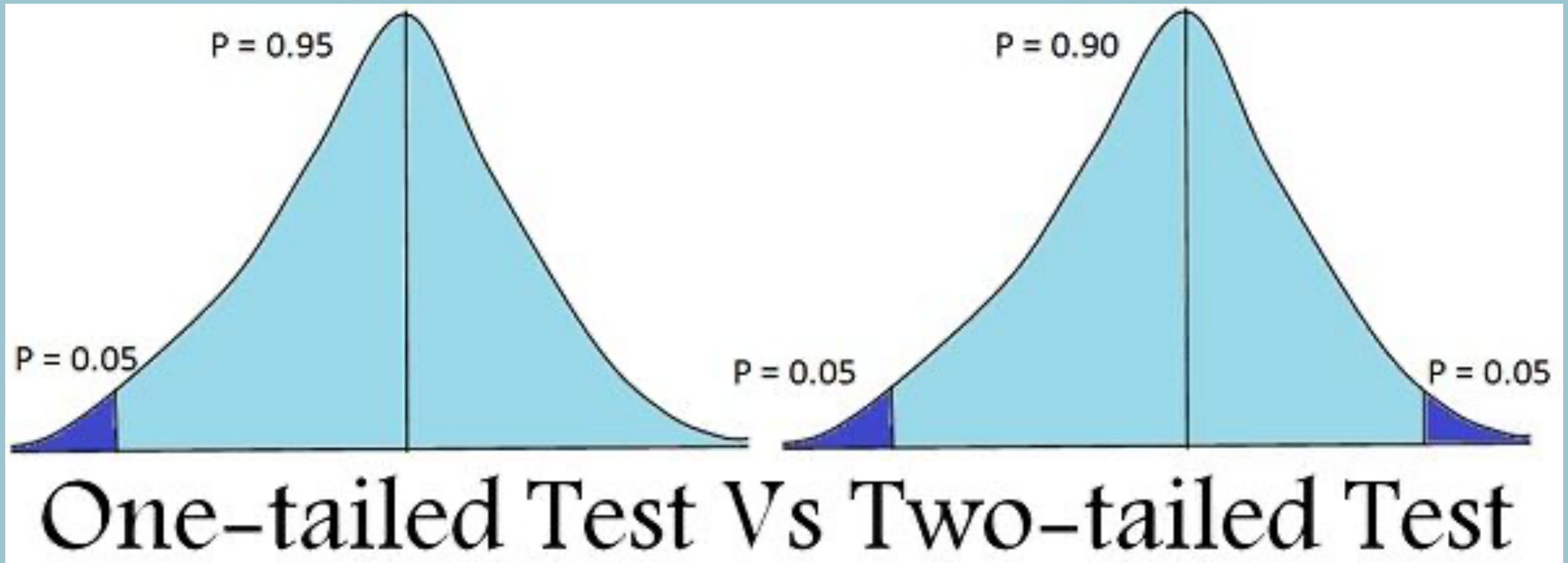
- One-tailed versus two-tailed test



Interlude:
Effect sizes

Cohen's d: $$d = \frac{\bar{X}_2 - \bar{X}_1}{s}$$

with $$s = \sqrt{\frac{(n_1 - 1)\sigma_1^2 + (n_2 - 1)\sigma_2^2}{n_1 + n_2 - 2}}$$

Interlude: Effect sizes

- Hypothesis tests only say: “This result is unlikely under the null hypothesis”

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- Scale-free measure of signal versus the noise

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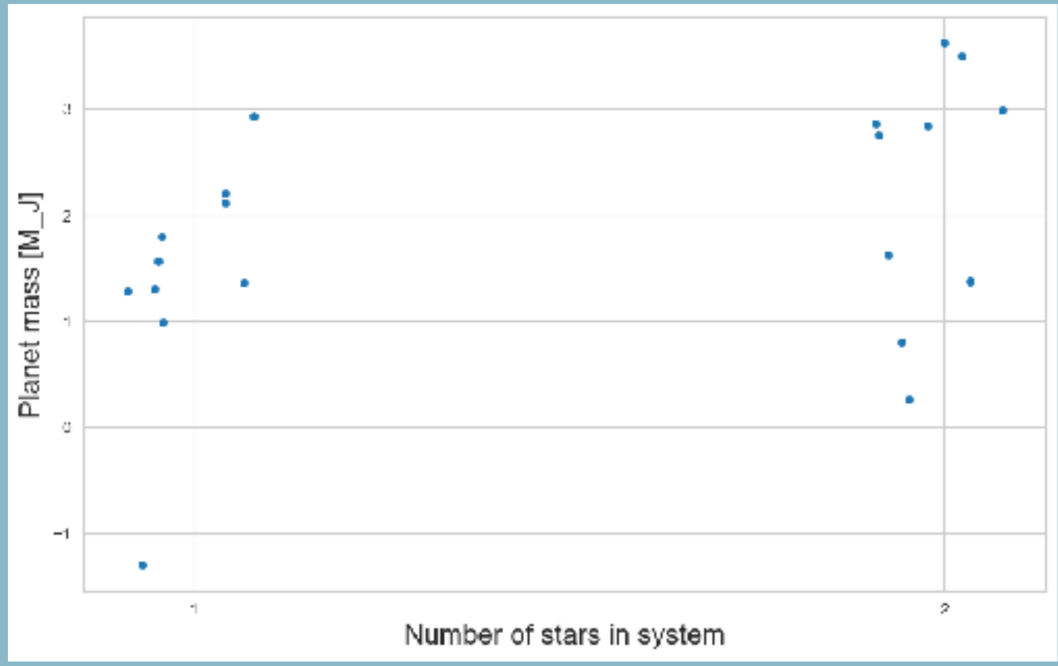
The Two-Sample t-test



$$T_t = \frac{\bar{X}_2 - \bar{X}_1}{\sqrt{\sigma_2^2/N_2 + \sigma_2^2/N_2}}$$

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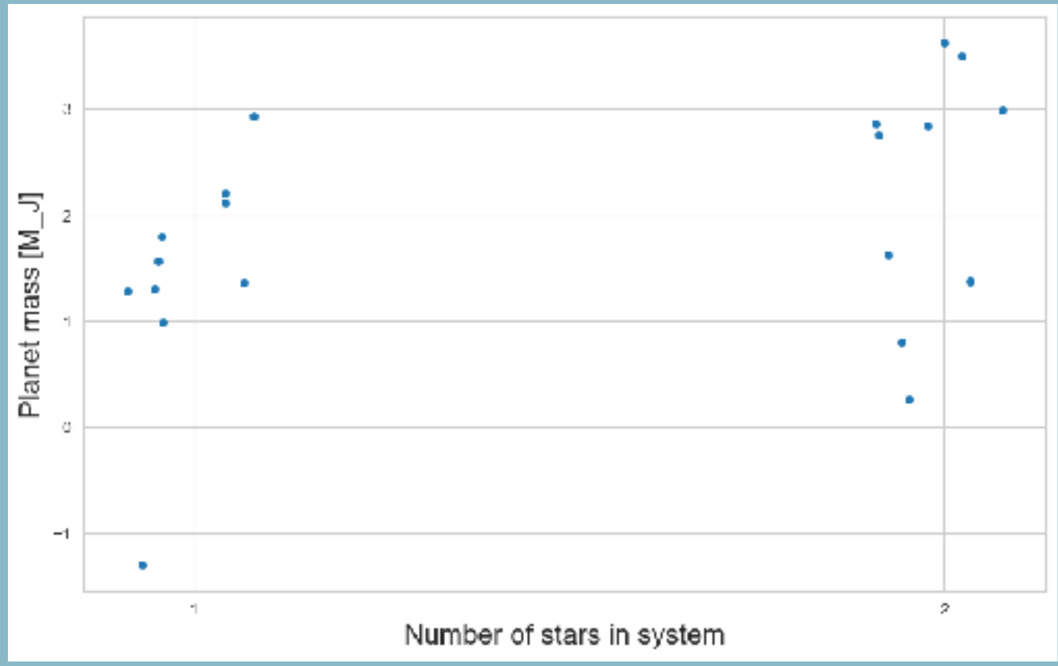

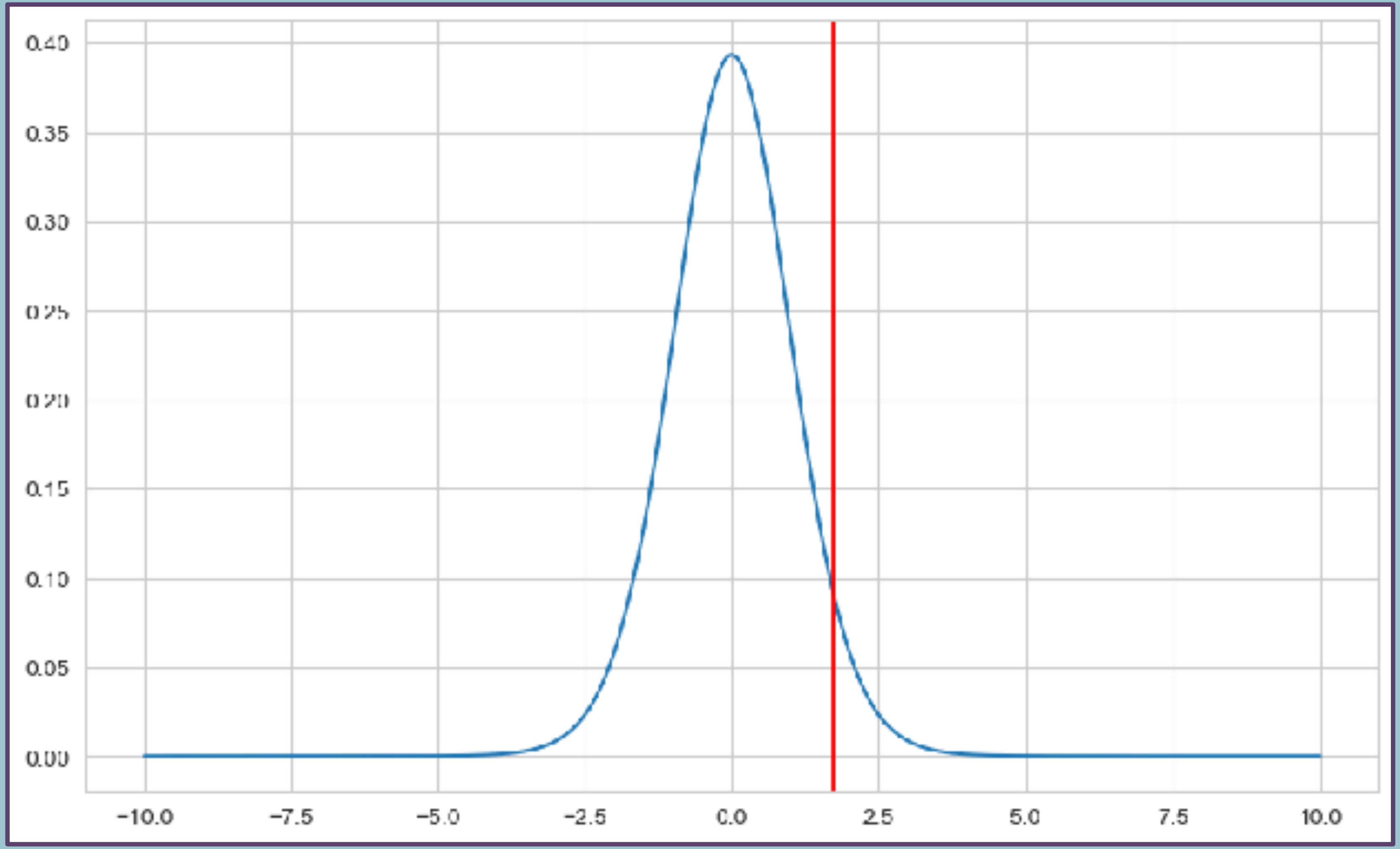

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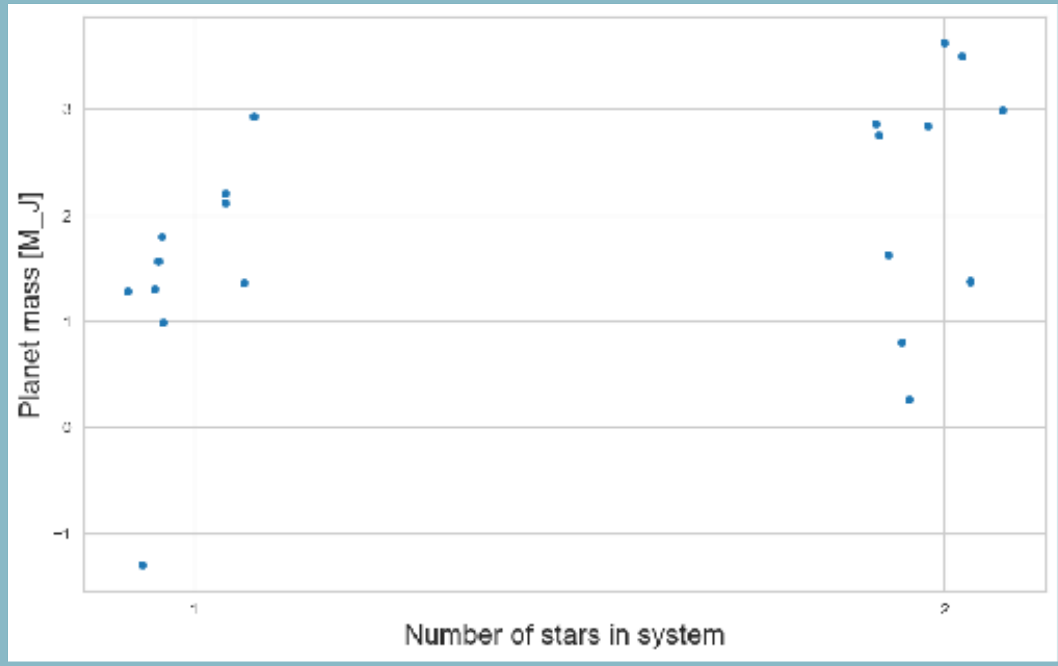

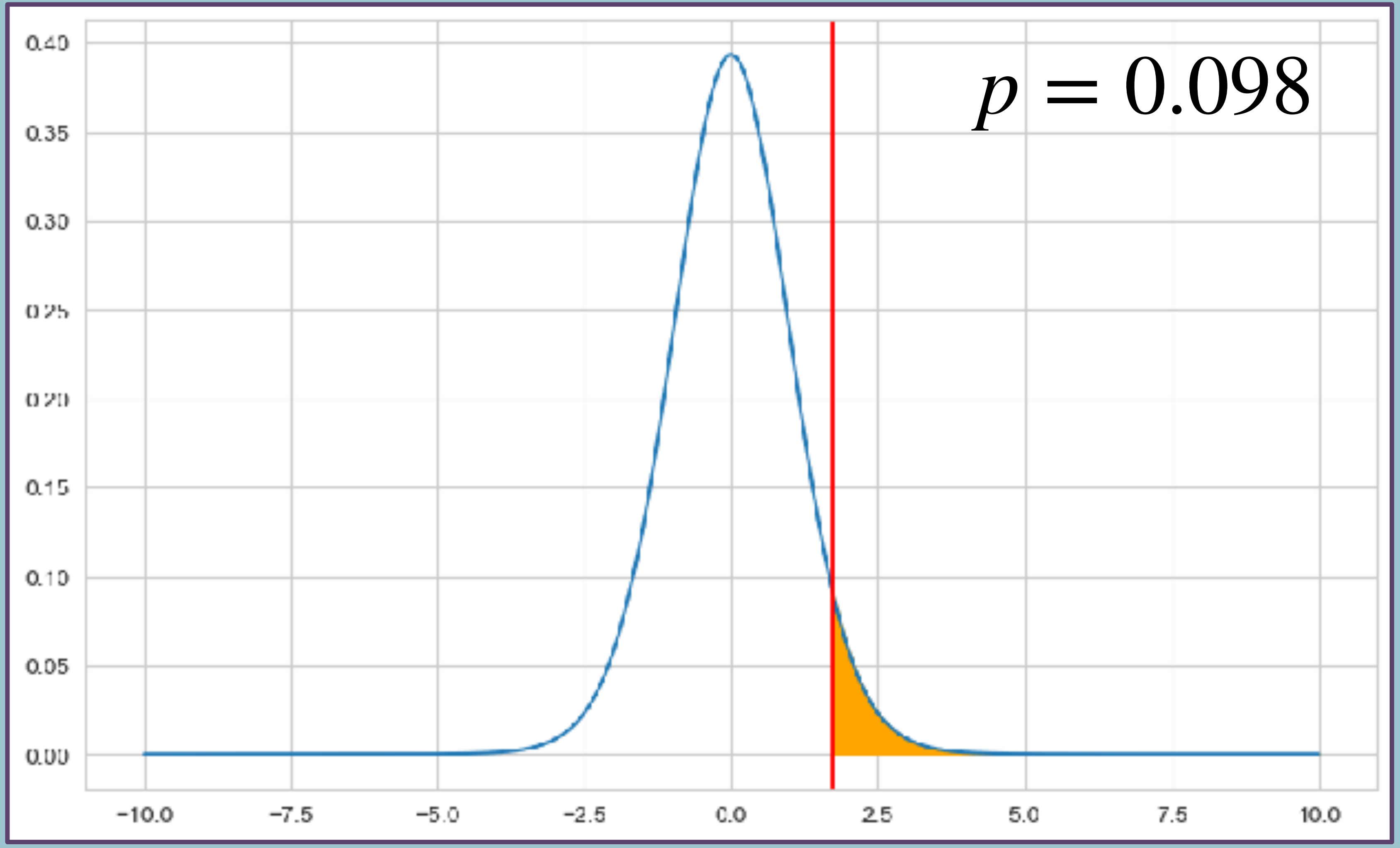

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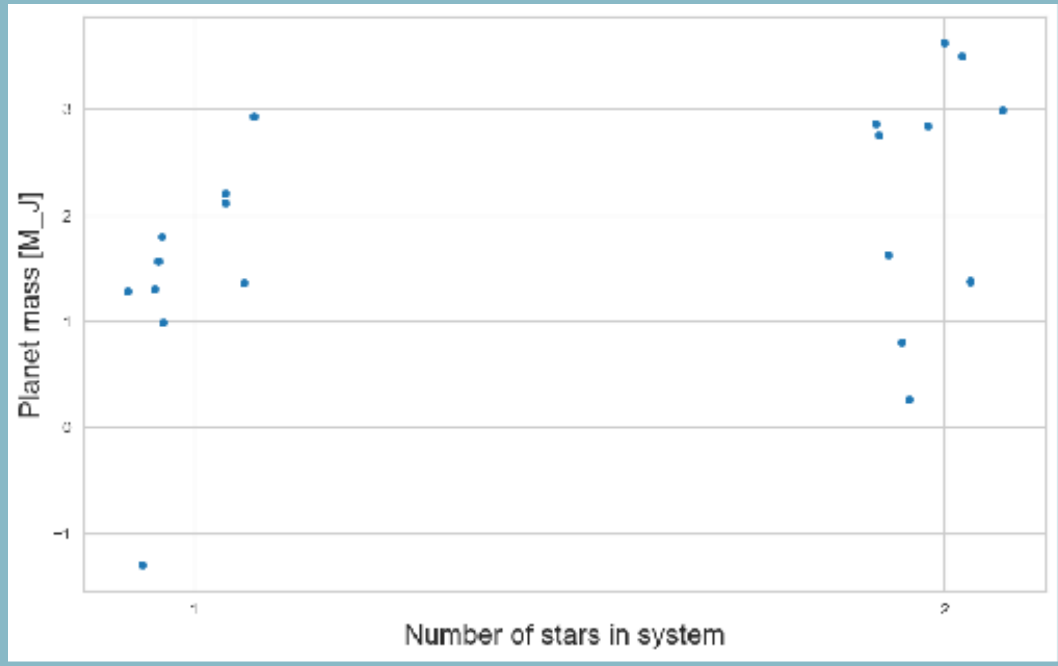

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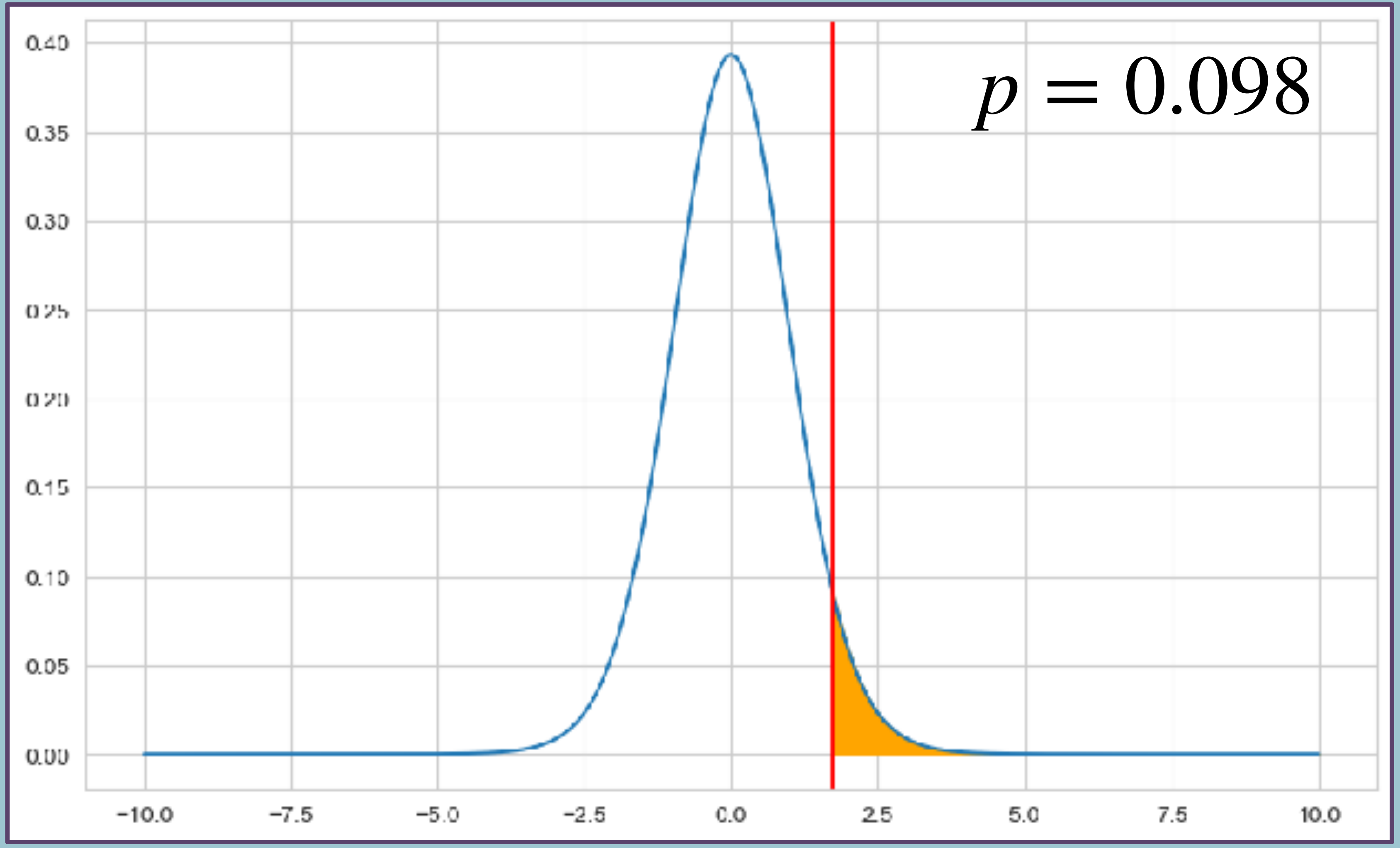
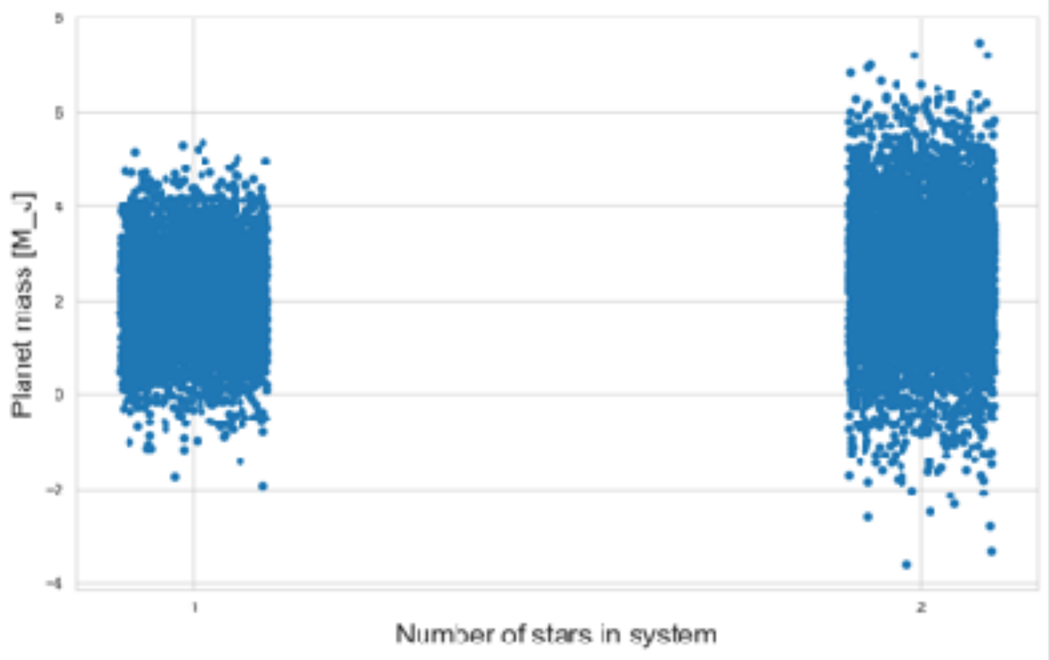


Fisher: p=0.098 is the significance of the test

Neyman-Pearson: set threshold at a critical value $T_{t,\mathrm{crit}}$, below you reject H0 and accept H1, above you do not reject H0

The Two-Sample t-test



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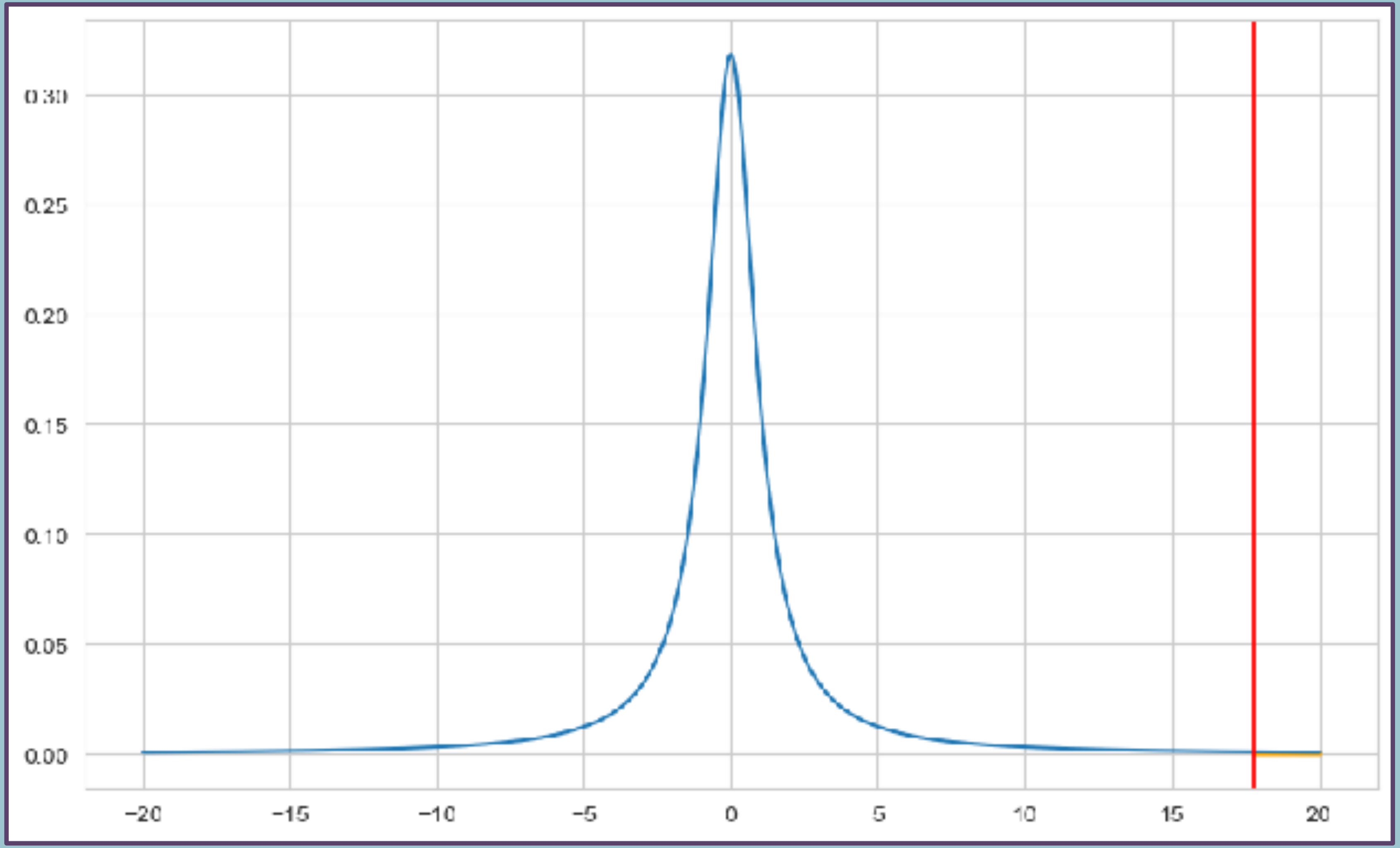

<u>Fisher</u>: p=0.001 is the significance of the test

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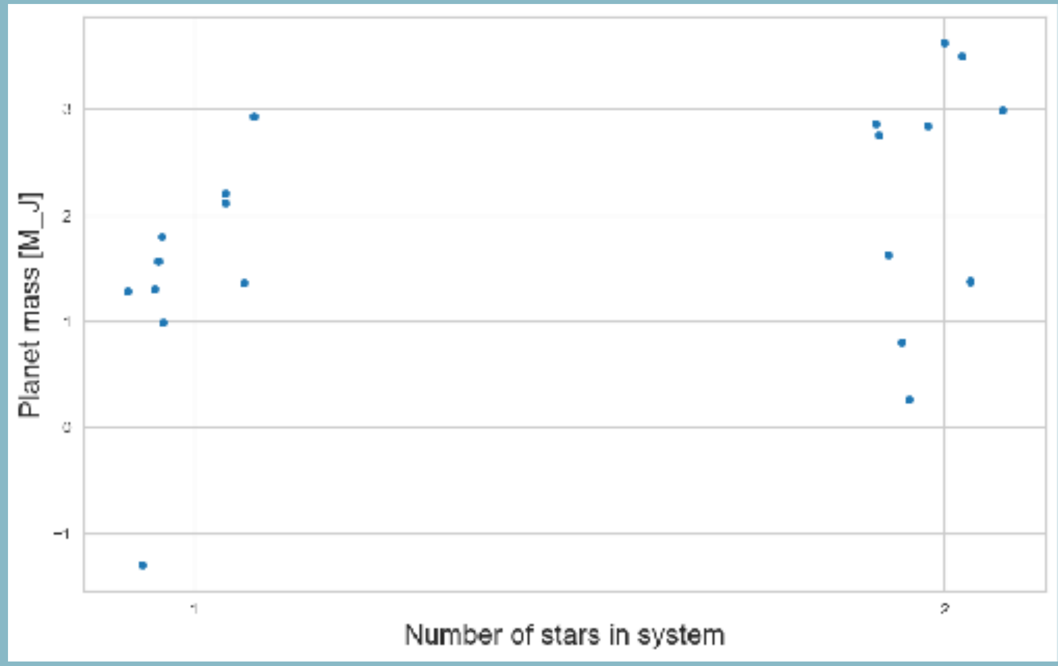
Hypothesis Testing



Conjecture:

Null hypothesis:

Test statistic:

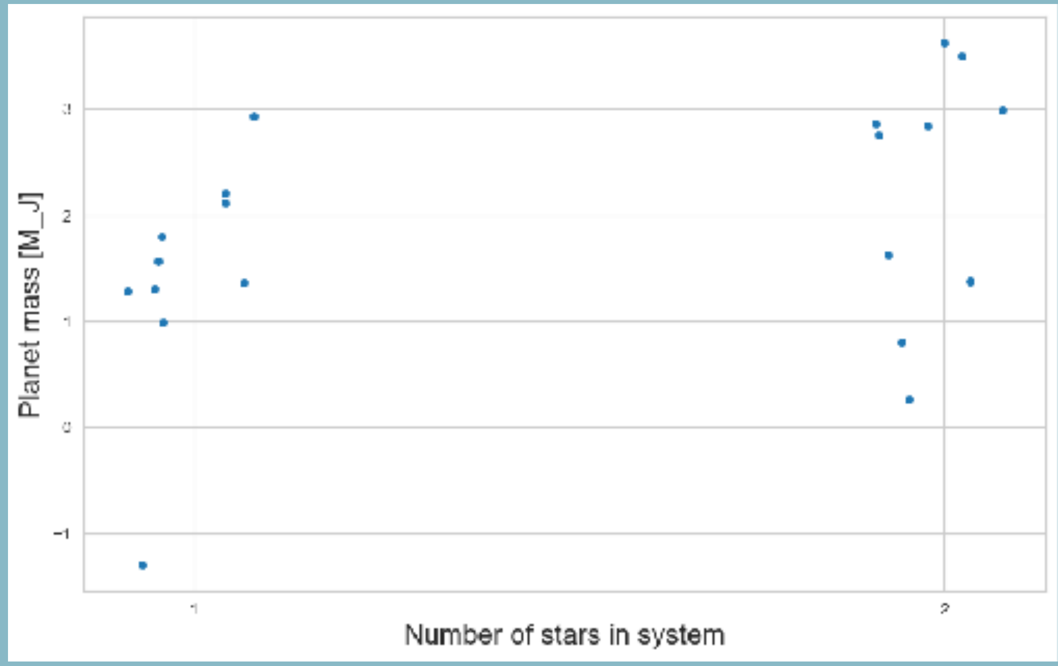
Statistical test:

Assumptions:

Test distribution:

Decision rule:

Hypothesis Testing



Conjecture: binary star systems produce more massive planets

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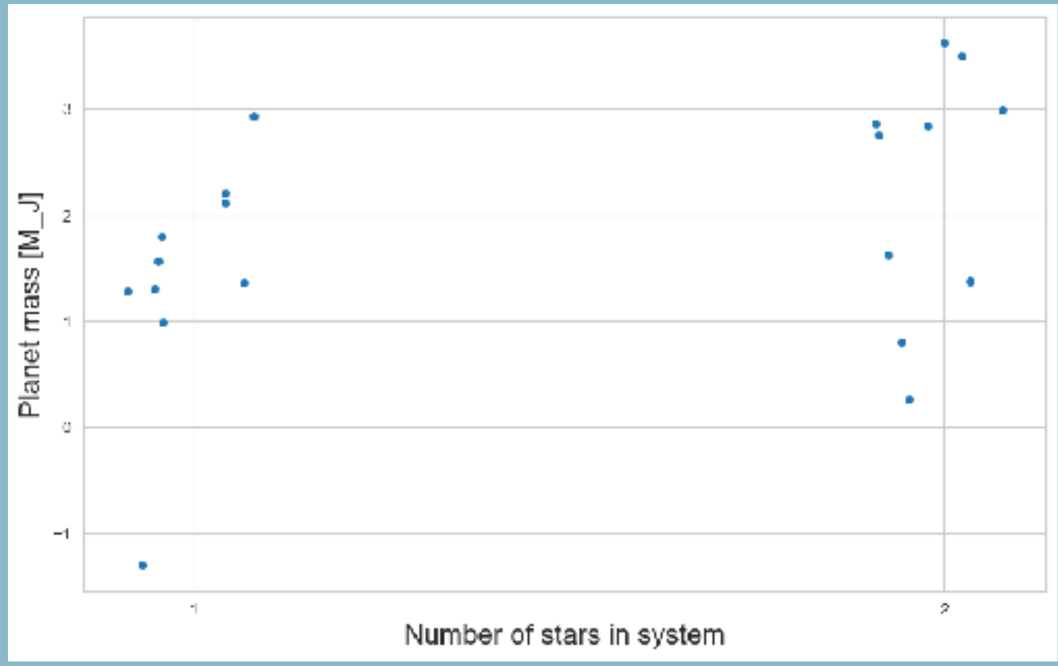
Hypothesis Testing



Conjecture: binary star systems produce more massive planets

Null hypothesis: the mass distributions around single and binary stars are the same

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Assumptions:

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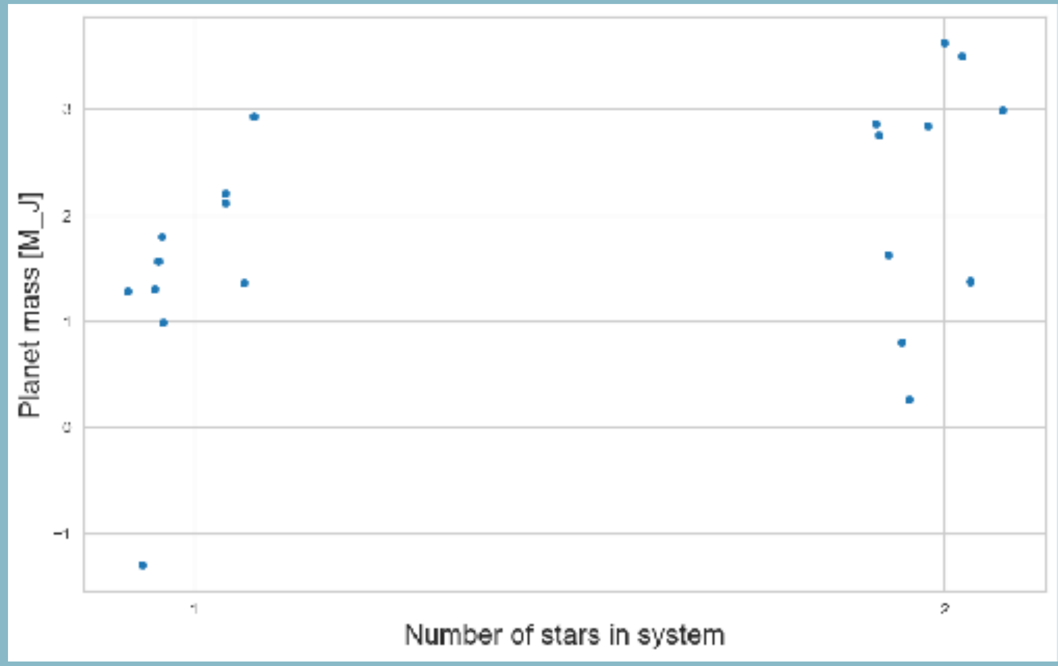

<u>Conjecture</u>: binary star systems produce more massive planets

<u>Null hypothesis</u>: the mass distributions around single and binary stars are the same

<u>Test statistic</u>: difference of means

<u>Statistical test</u>:

<u>Assumptions</u>:

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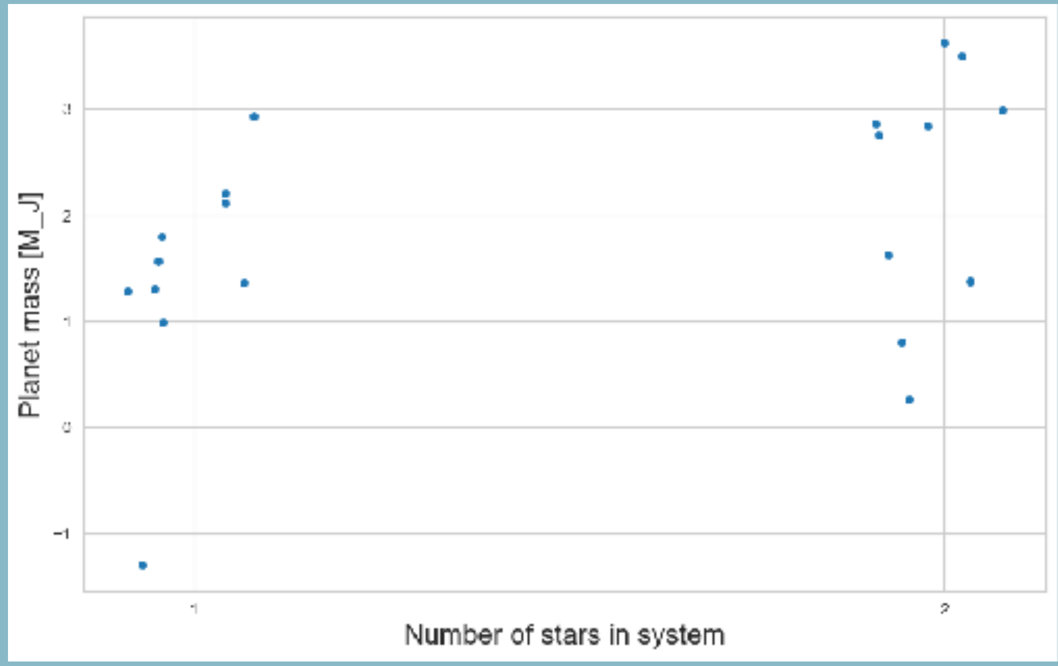
Hypothesis Testing



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<u>Statistical test</u>: t-test

<u>Assumptions</u>:

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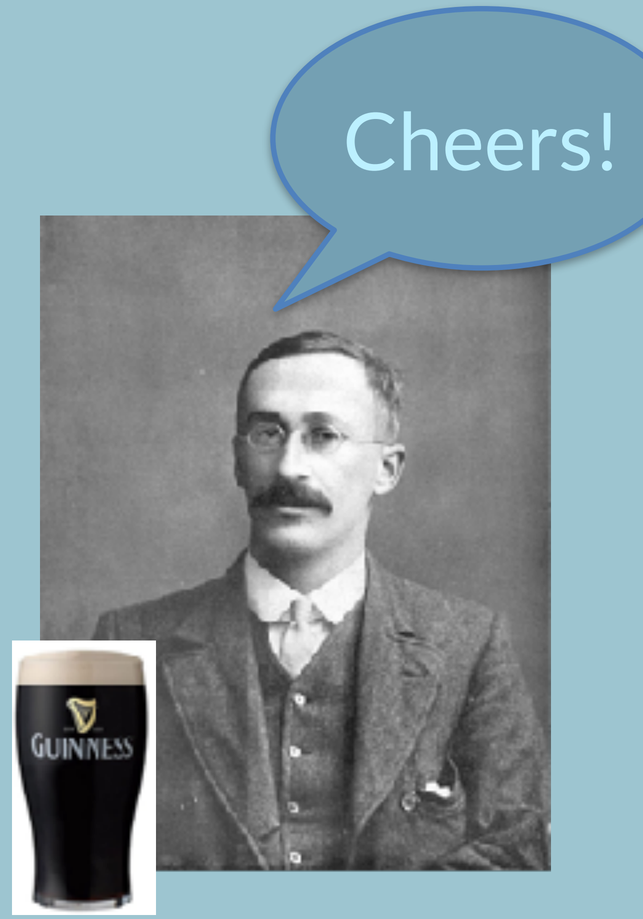
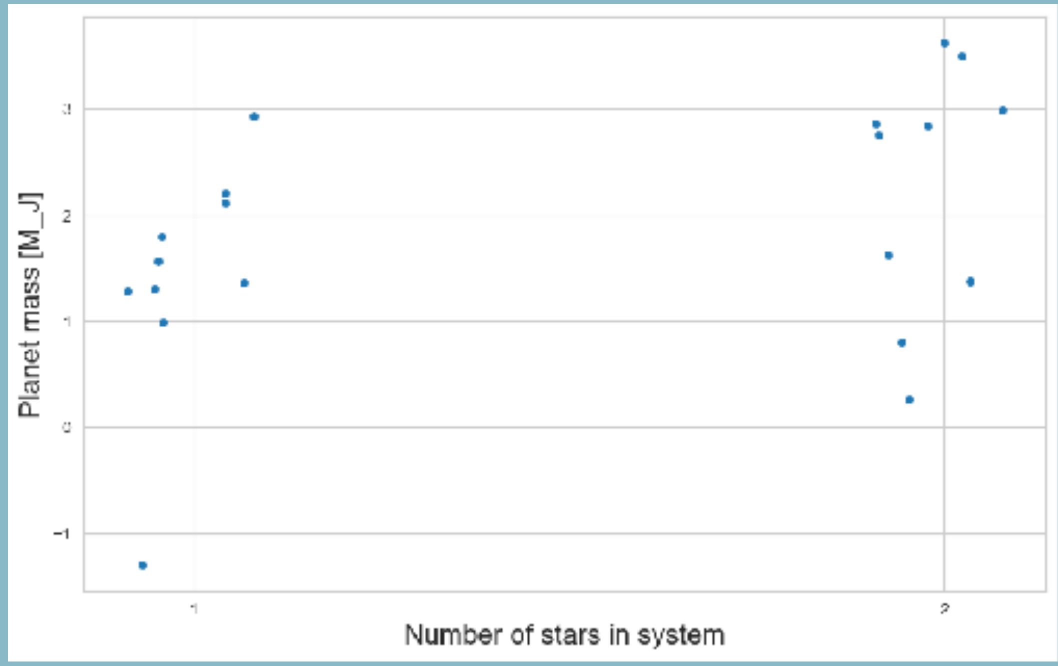
<u>Decision rule</u>:



Hypothesis Testing



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<u>Null hypothesis</u>: the mass distributions around single and binary stars are the same

<u>Test statistic</u>: difference of means

<u>Statistical test</u>: t-test

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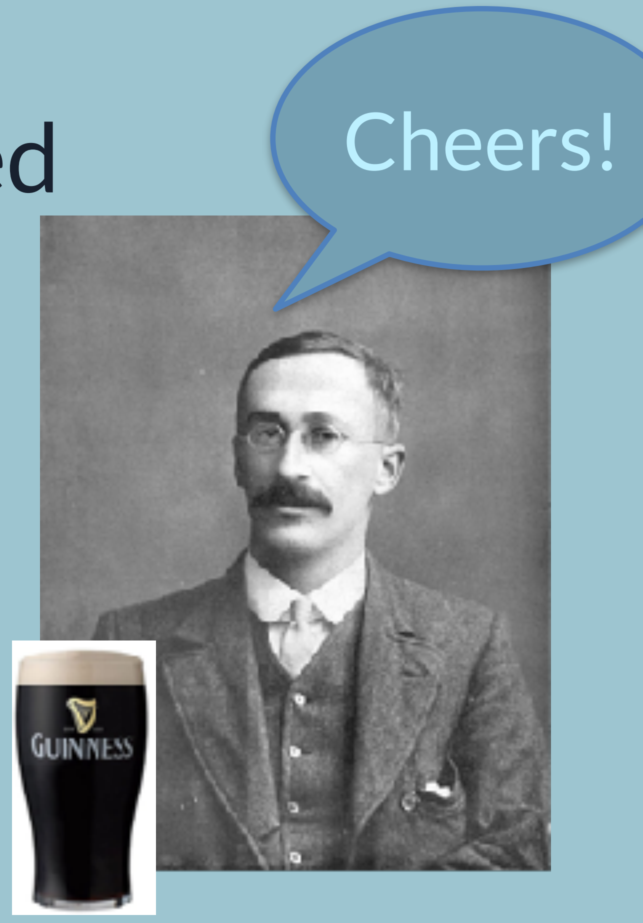

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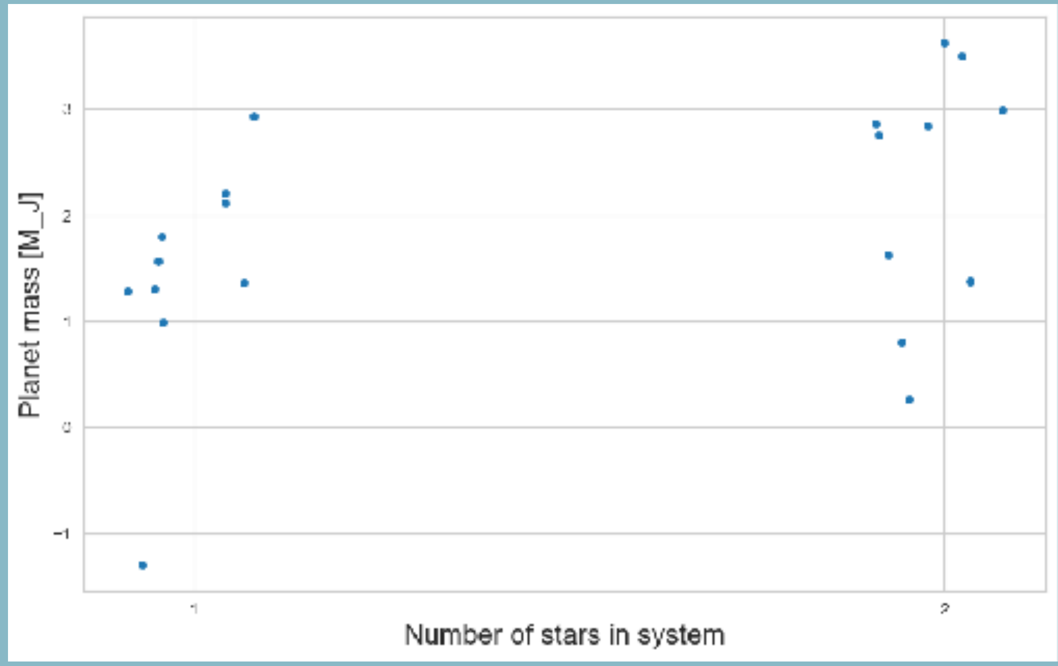

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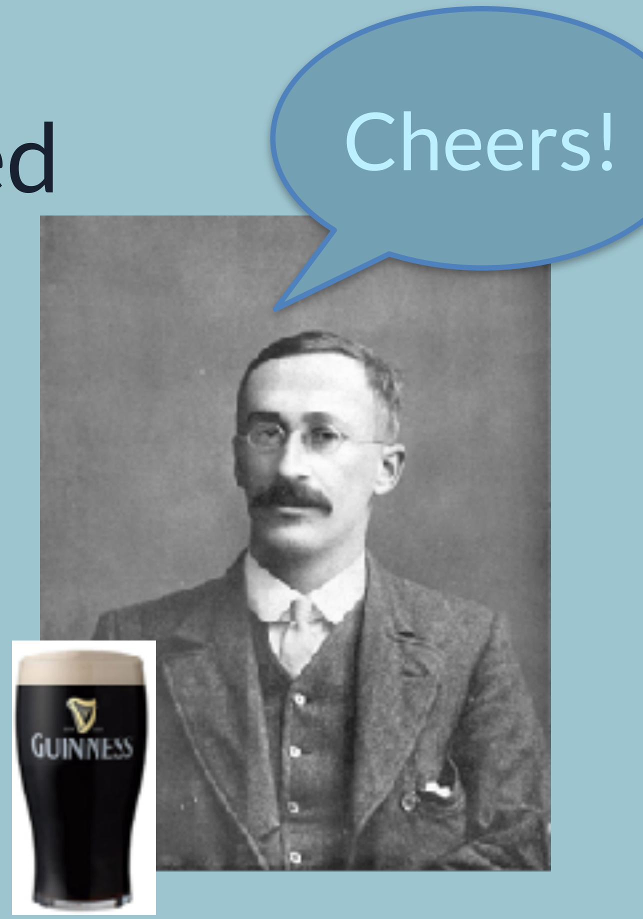

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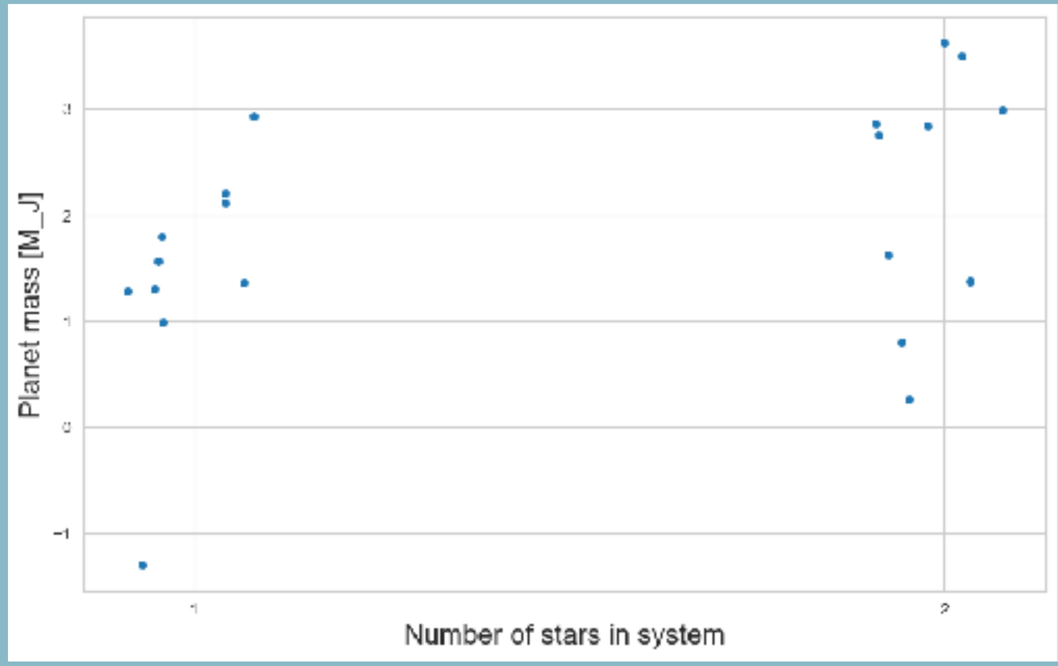

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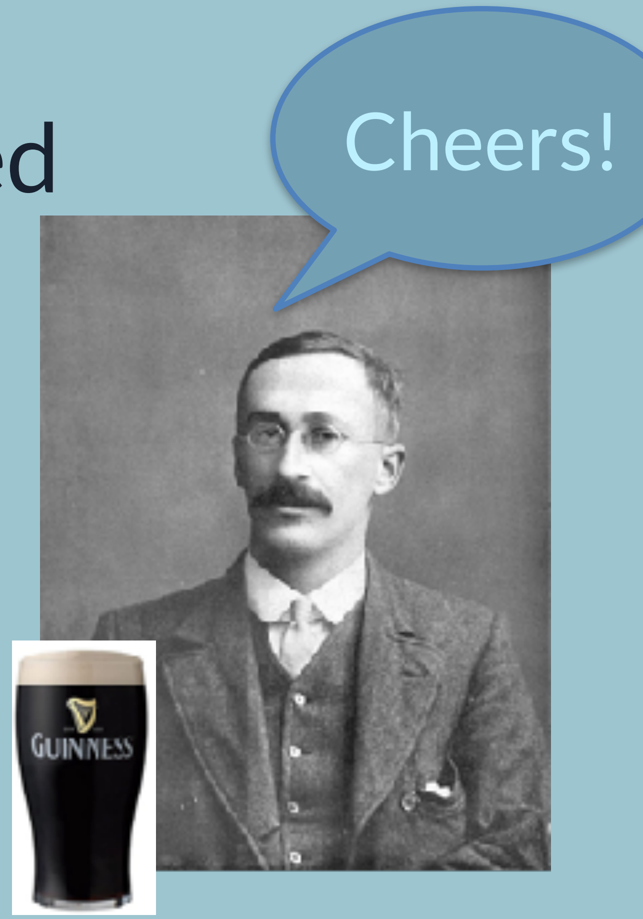

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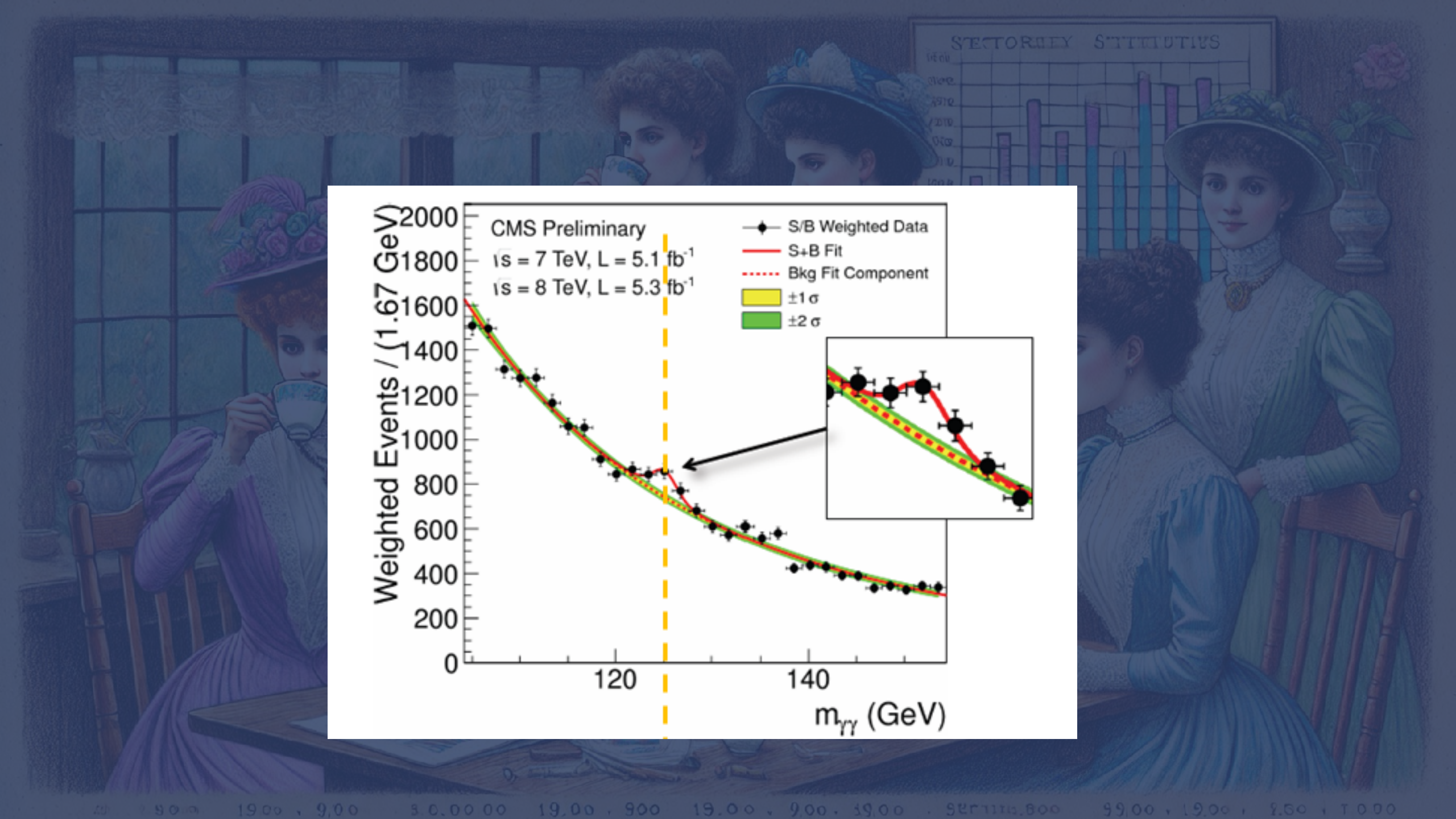
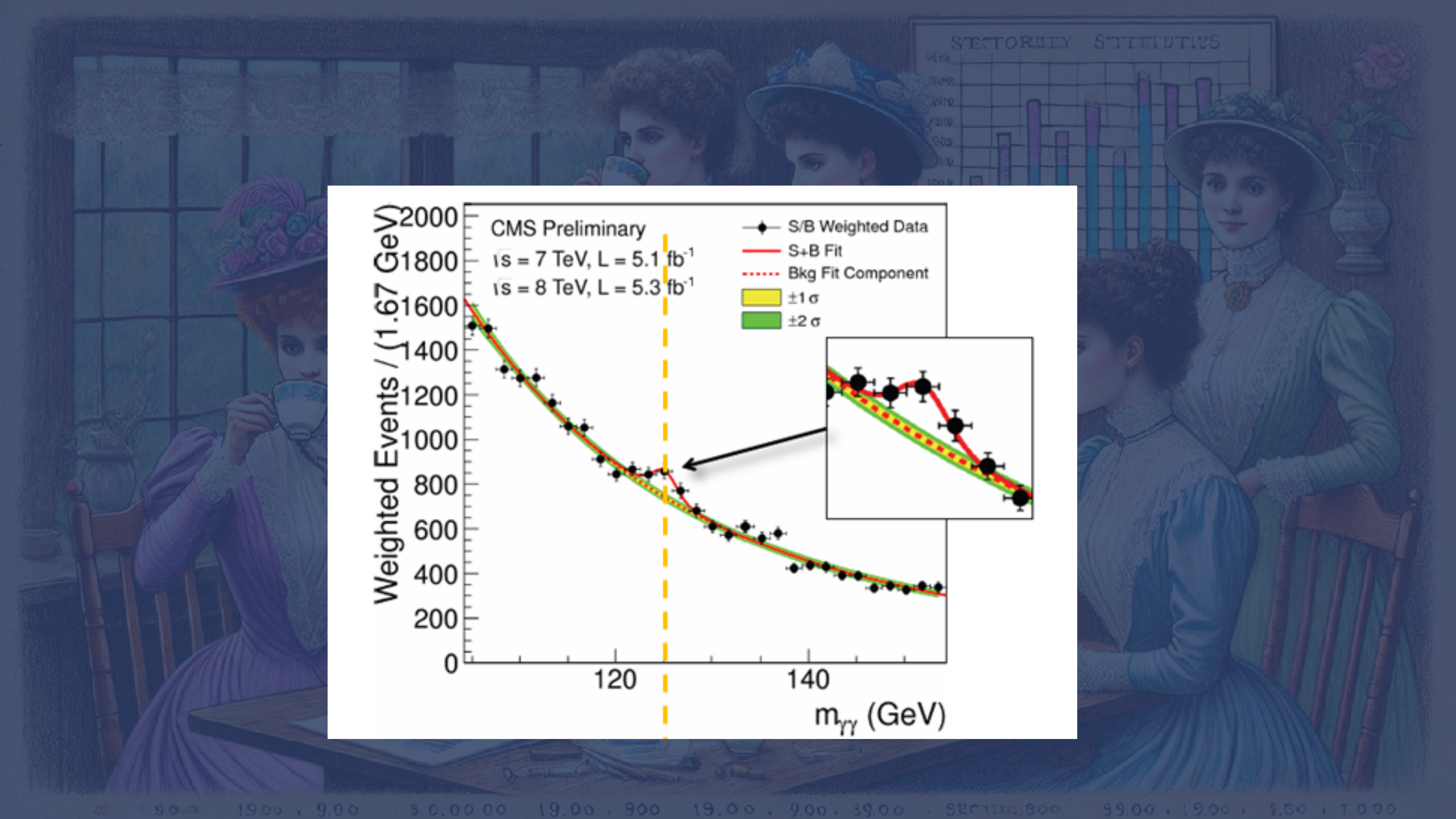
- Not rejecting the null hypothesis is also <u>not direct evidence for the null hypothesis</u>. It just means you do not have enough information to reject it.


CMS Preliminary
√s = 7 TeV, L = 5.1 fb-1
√s = 8 TeV, L = 5.3 fb-1
S/B Weighted Data
S+B Fit
Bkg Fit Component
±1 σ
±2 σ
Weighted Events / (1.67 GeV)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
120
140
mγγ (GeV)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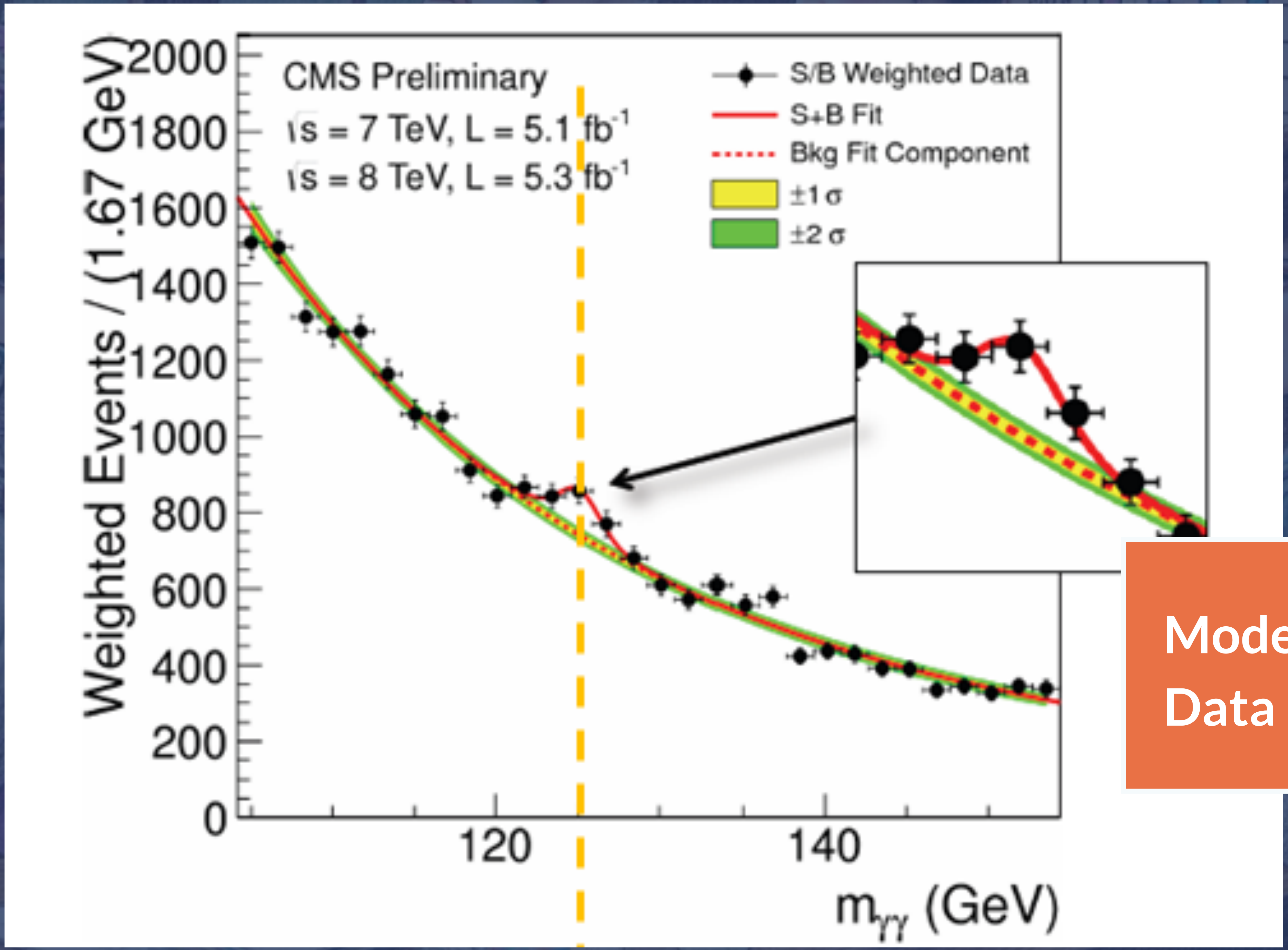
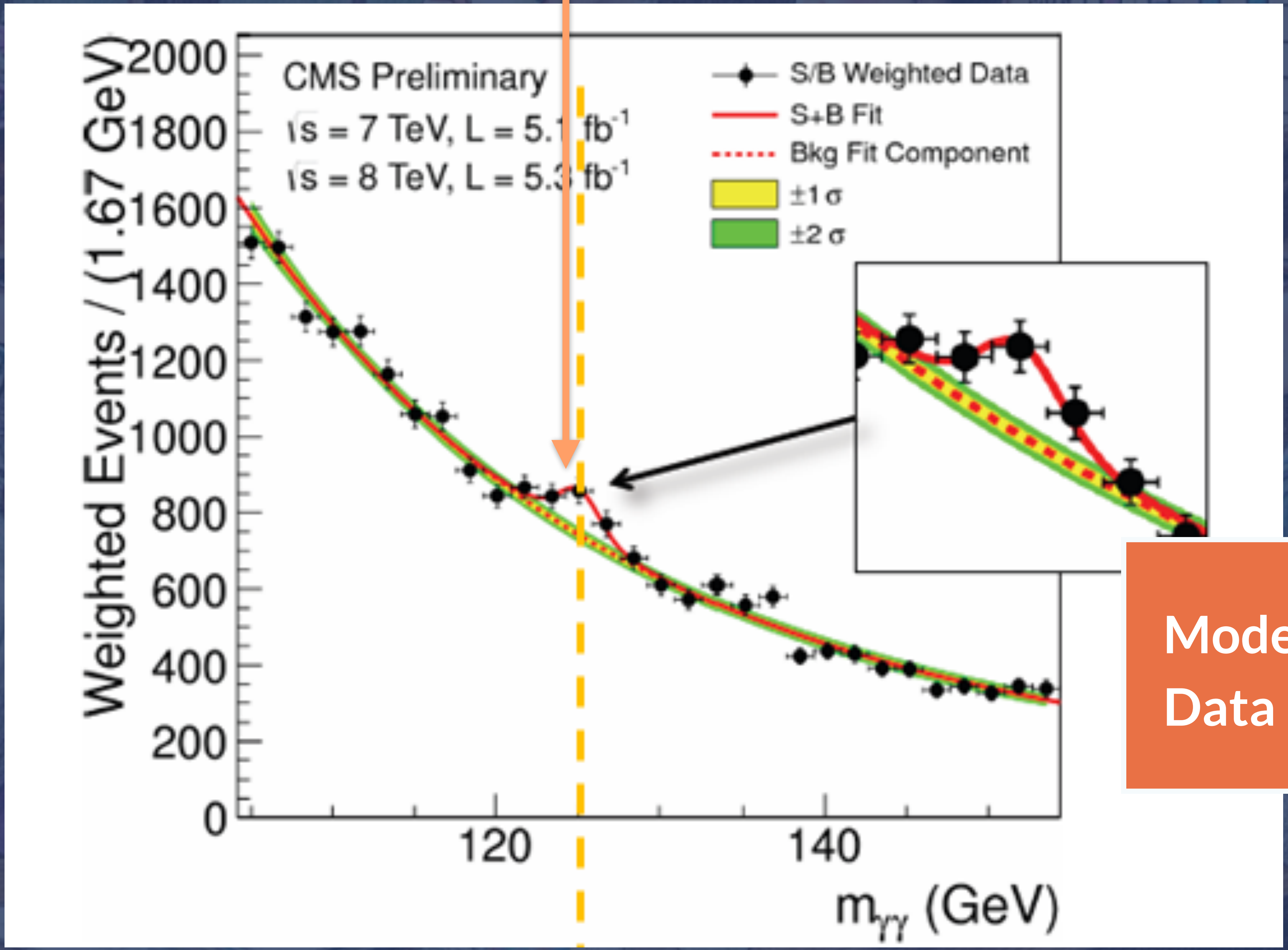
2000
1800
1600
1400
1200
1000
800
600
400
200
0
120
140
mγγ (GeV)



Model misspecification?

Data issues?

Is there a new particle in the CMS data?



Model misspecification?
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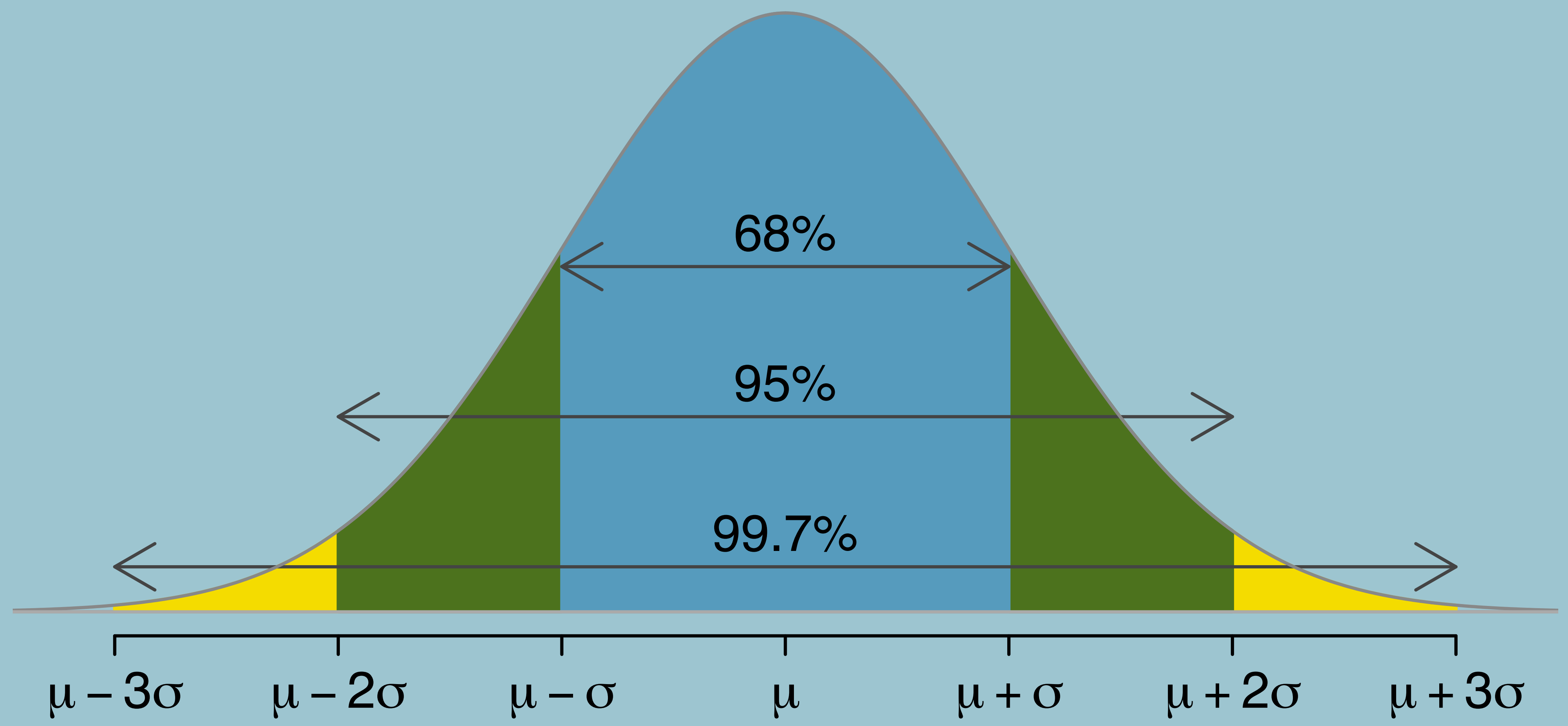
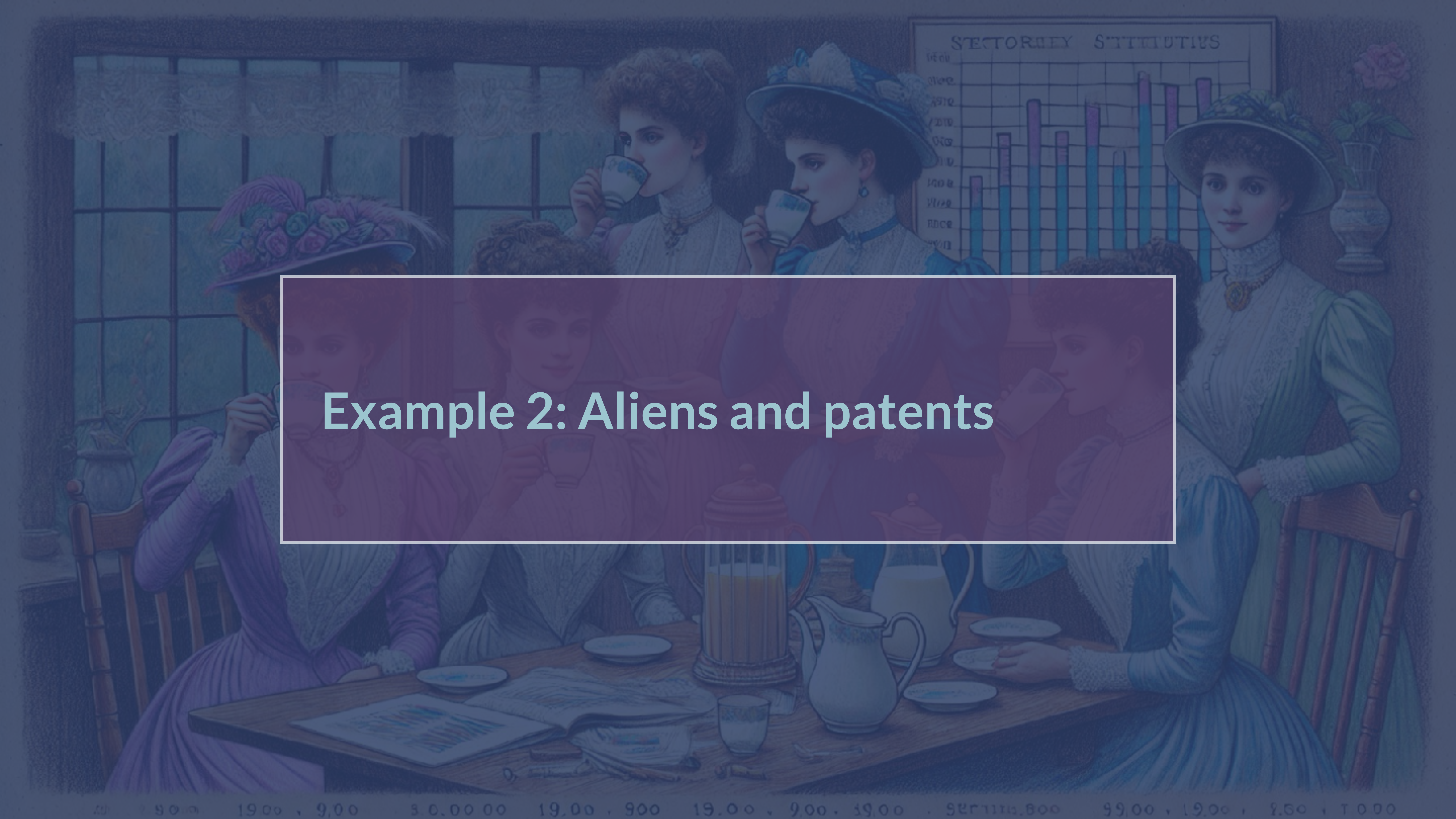
TO FINISH: Add Higgs Boson example

Hypothesis tests and sigmas (revisited)

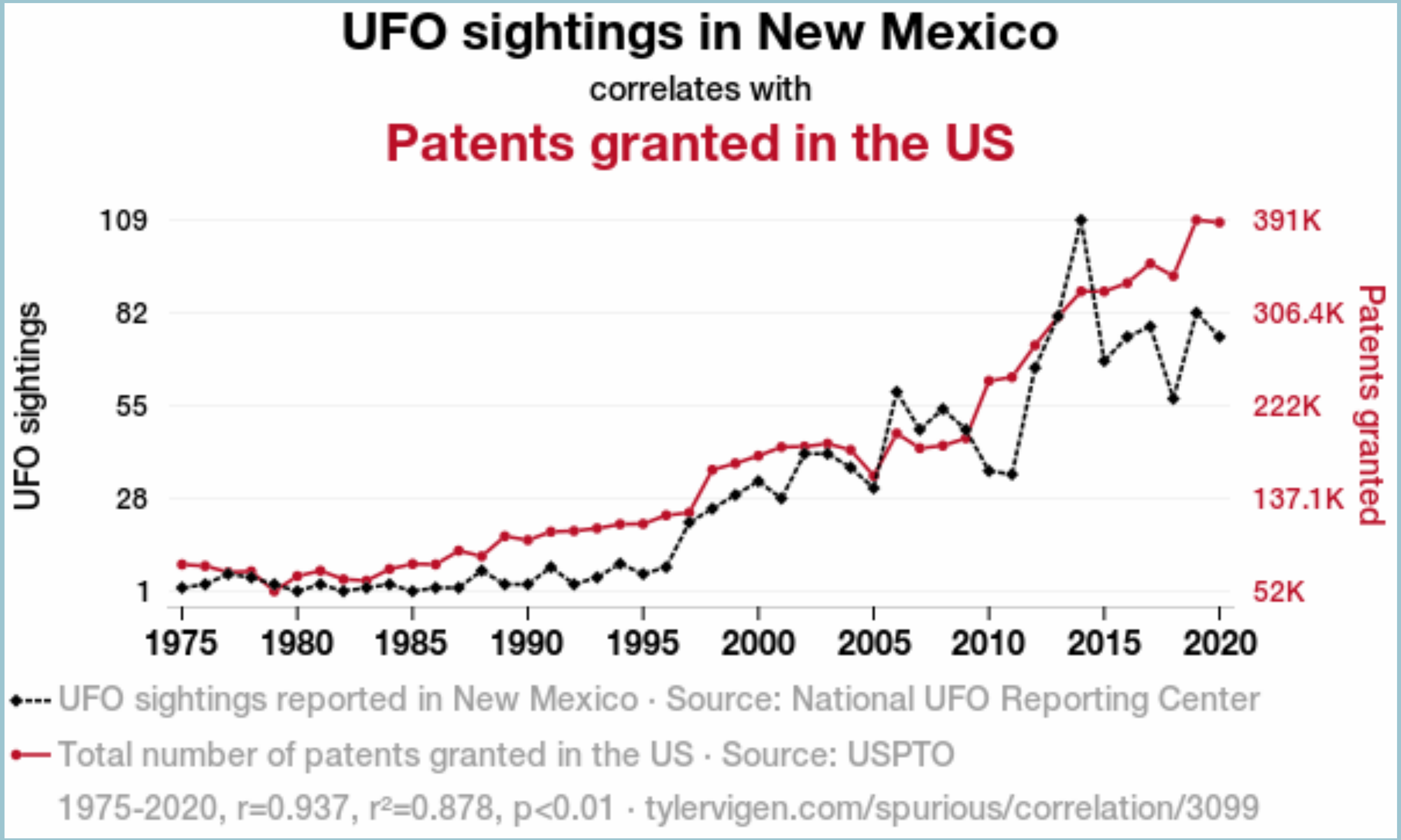




Example 2: Aliens and patents

Are most US patents submitted by aliens?

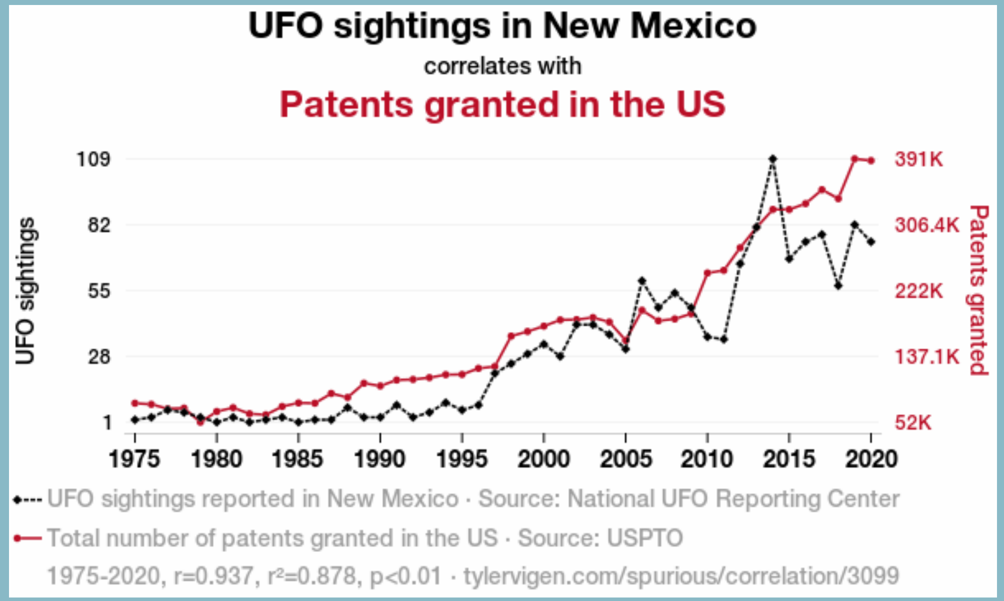
"The influx of intergalactic inspiration in New Mexico led to an explosion of creative scientific ideas, spurring a wave of groundbreaking inventions and technological advancements. It turns out, alien innovation is the real secret behind some of our greatest patented achievements!"



What is your conjecture and null hypothesis?

What quantity could we calculate to test this hypothesis?

Correlations

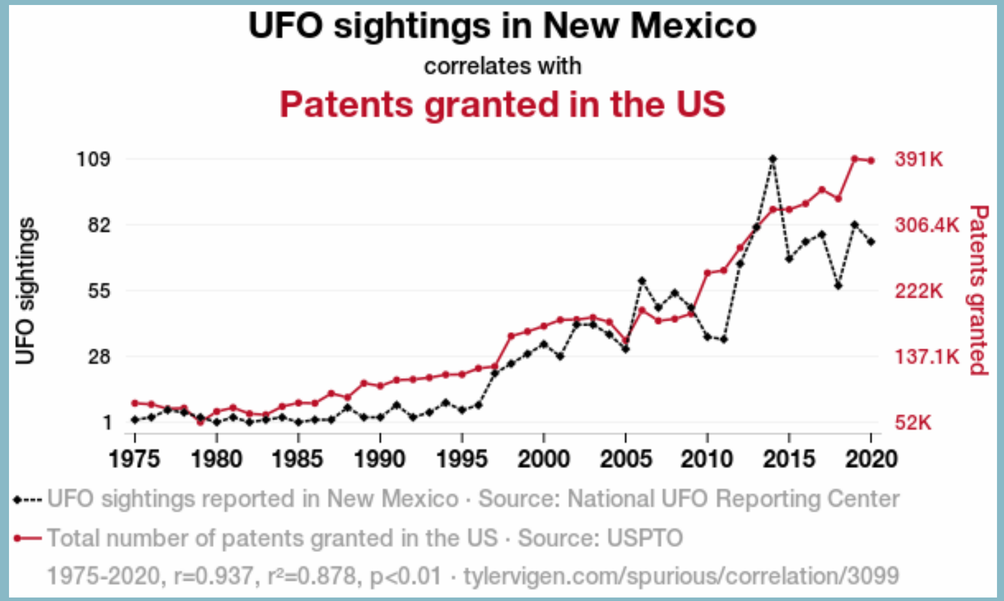


Sample covariance:

Correlation coefficient (Pearson's r):

Spearman's ρ:

Correlations



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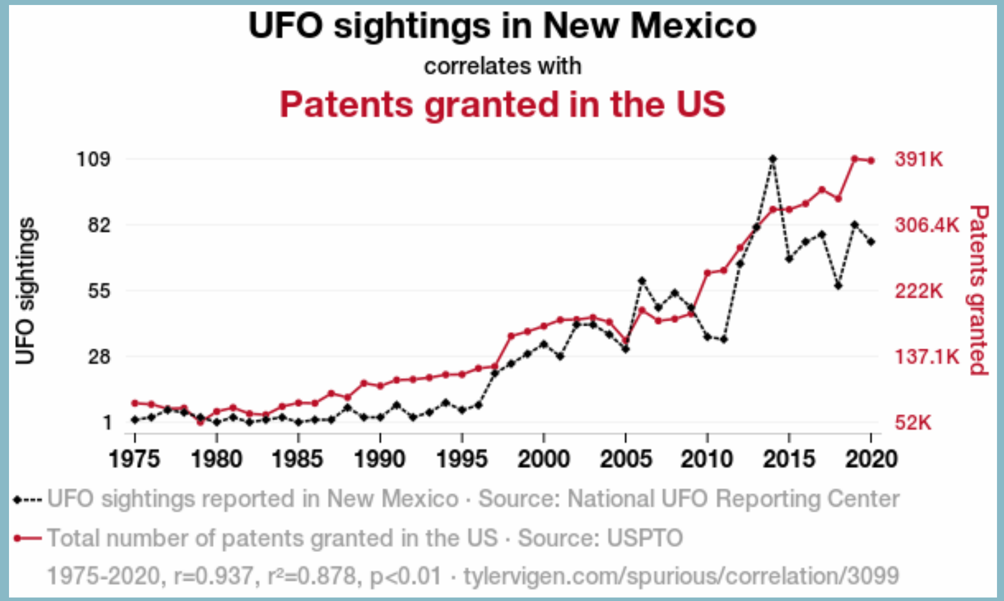

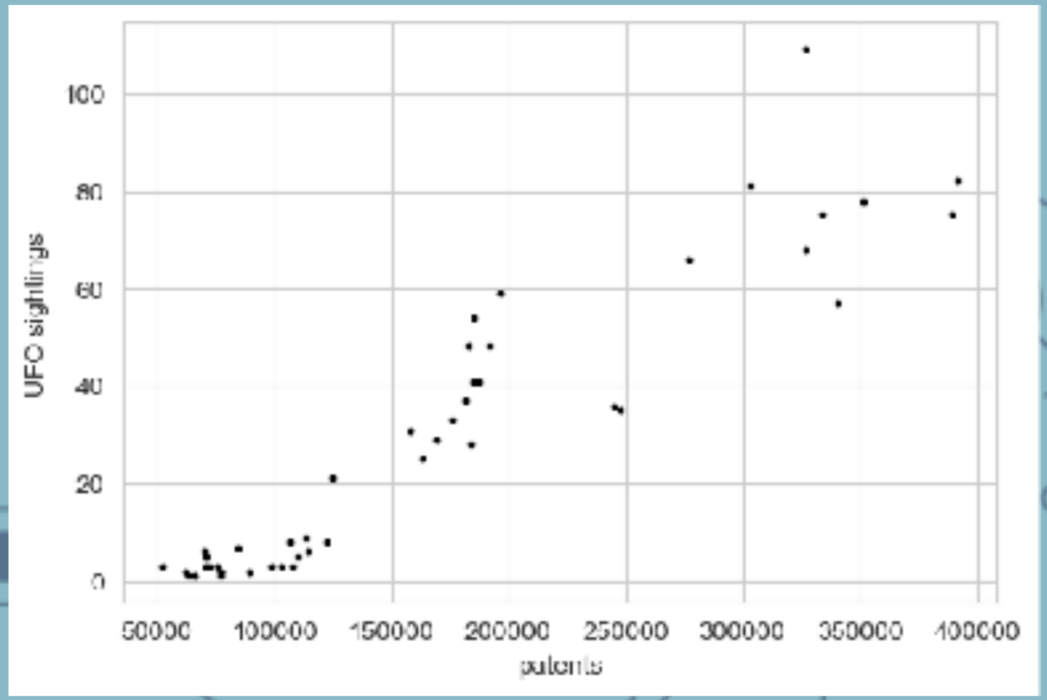

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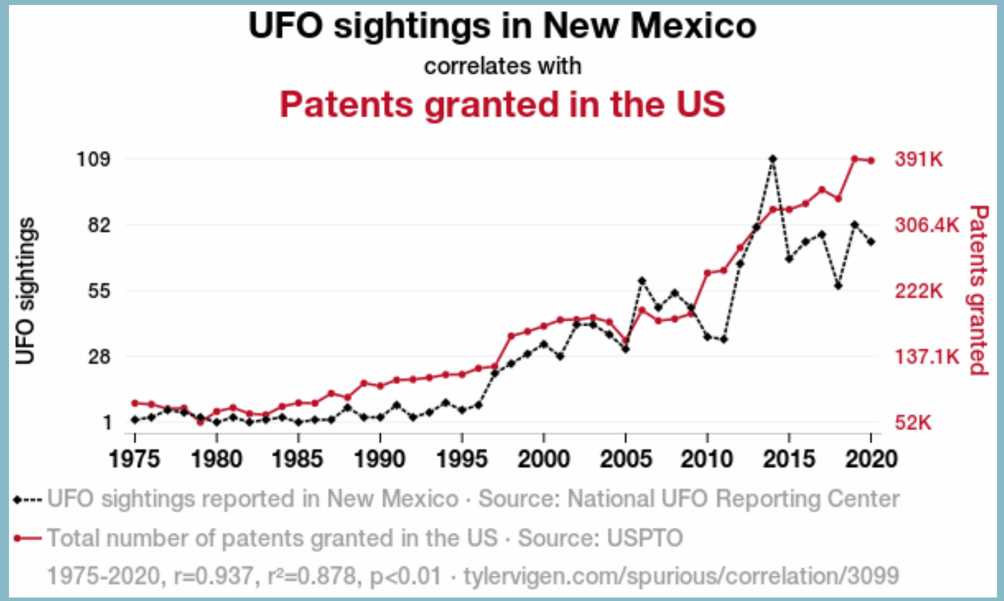
Alien patents



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Null hypothesis:

Test statistic:

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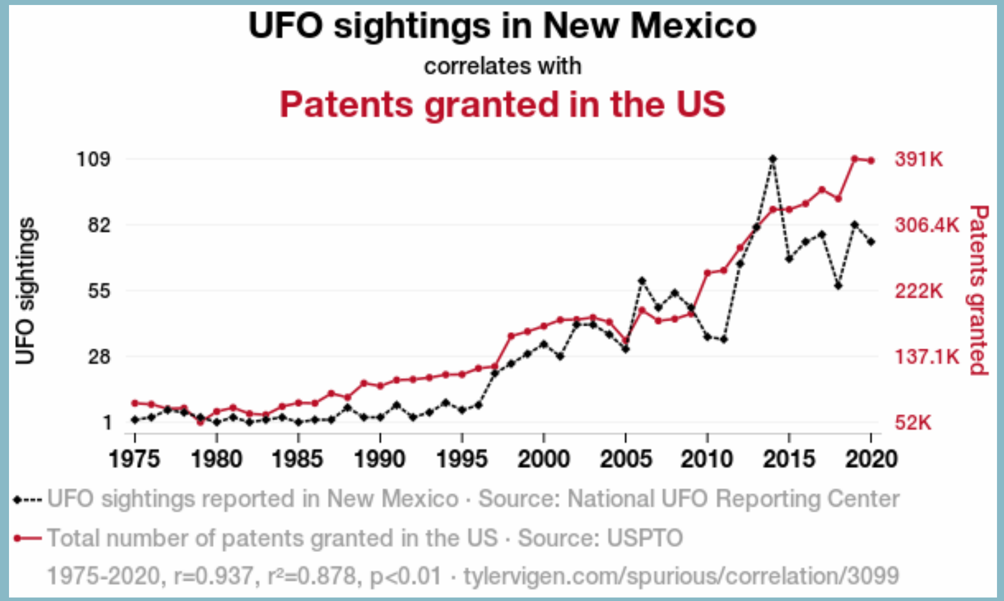

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Result: $r=0.937$, $p<0.01$

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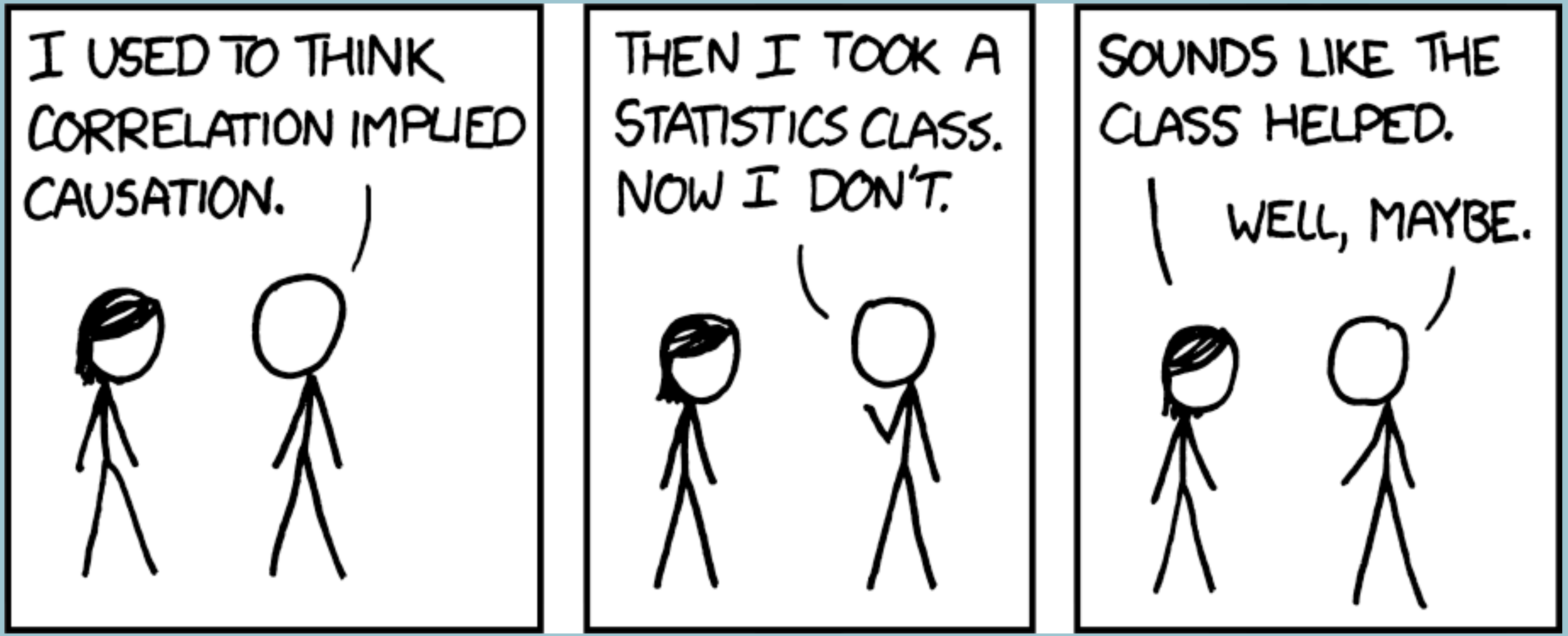

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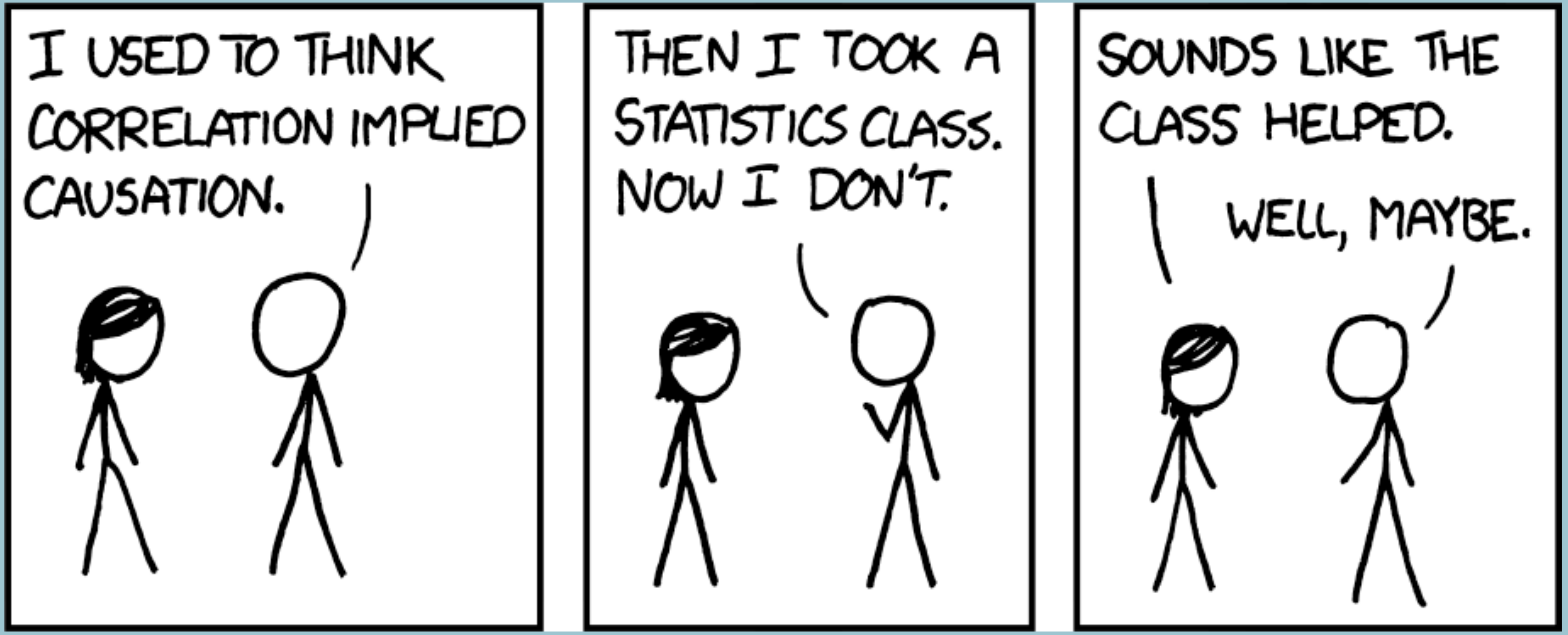

Note that causation does imply correlation!

Issue 2: Number of trials

Database with **25237** variables, **636.906.169** correlations total.

How many significant correlations would we find, by pure chance, if we only considered correlations at $p<0.01$?

What threshold would we need to set in order to only consider correlations significant such that we have a 1% probability of obtaining one per chance across the entire dataset?

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If you do multiple hypothesis tests, take the number of trials into account when computing significance!